Abstract

"Determine the Best Practices and Strategies for Reopening Bellevue College Safely in the Fall under COVID-19 Condition: A Recommendation Report"

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With areas in the United States reopening schools and businesses, and the COVID-19 situation still persisting, it was necessary to look at how this would affect Bellevue College. On May 10, Dr. Robert Viens approved our proposal to gather information on what it would look like if students were to return to campus and the effects the virus would have on the health of students and faculty. In order to determine this risk, we had to analyze the danger of the virus, including its transmissibility. We looked into the plans Bellevue College has established regarding what to do if a student or faculty member contracted COVID-19. With the possibility of a return to campus not being viable for the fall quarter 2020, we researched what sorts of technologies would be required to assist students and teachers with online classes, including if any software, hardware, or Internet upgrades would be beneficial. To perform this research, we coordinated with Professor Peterson to gather information from Bellevue College. We found that although Bellevue College has a healthy population compared to the general public, COVID-19 still presents a significant risk to students and faculty, especially those with underlying health conditions, and therefore extensive measures to keep the virus from spreading must be taken. In the meantime, there are technological upgrades Bellevue College should consider to provide an optimal online learning environment. This report contains numerous recommendations regarding what actions Bellevue College should take to ensure the health and safety of faculty and students.

Keywords: coronavirus, COVID-19, campus return, online classes, student health, faculty health

Determine the Best Practices and Strategies for Reopening Bellevue College Safely in the Fall under COVID-19 Condition: A Recommendation Report

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Executive Summary

Due to the risk presented by COVID-19, many businesses, parks, schools, and other places in Washington state closed and a stay-home order was issued to slow the transmission of the virus. On May 10th, our team received approval from Dr. Robert Viens, Assistant Vice President for Academic and Student Affairs, to research the best practices and strategies for reopening Bellevue College safely and brainstorm different technology and programs that would help BC provide a more efficient instruction remotely during the lockdown period. In this short summary, we discuss the tasks we needed to research in order to obtain the necessary information and help BC make the best decisions. The format for these short summaries consists of the “problem”, the “results”, and the “conclusion and recommendation”.

Task 1

The goal of task 1 was to “acquire and understand what the health risks for campus employees and students are, based on the best, currently available information.” To determine the health risk we looked at articles, studies, and other available information on COVID-19. While studying the information we focused on both local, national, and international information on COVID-19. Our research revealed that COVID-19 is a virus that results in mild to severe respiratory illness. COVID-19 is mainly transmitted from person to person by way of contact, through emitted respiratory droplets. Though this seems to be the main vehicle of virus transmission, a person can also become infected by touching a contaminated surface and aerosols. Typical symptoms of COVID-19 include fever, cough, shortness of breath, and loss of taste and smell. Though certain symptoms are typical, the virus can also produce asymptomatic reactions, where a person is infected but does not show any symptoms. Globally, as of May 19th, there have been around 5 million COVID-19 infections, of which 300,000 resulted in death. Interestingly, we found that COVID-19 is more harmful to older people, and is a serious threat to individuals over 60 years old. Our research also showed that the prevalence of COVID-19 changes depending on the community, with those of Native Hawaiian, Pacific Islander, Hispanic, and African American descent being most at risk. Despite efforts to produce effective medicinal treatment, currently, there are no vaccines or cures for COVID-19. Due to the incurable nature, social distancing is currently the most effective method to prevent infection. With that being said, we concluded that COVID-19 poses a serious risk to Bellevue College faculty and students. Thus, we recommend that, until a vaccine is developed, students and faculty should avoid contact with others in order to

stay healthy. We also recommend that Bellevue College officials stay up to date with new information regarding the pandemic and adhere to local and national governmental guidance.

Task 2

The goal of task 2 was to “obtain data on the demographics of the students and campus employees to better assess their health risks.” To accomplish this task we contacted Professor Peterson, who provided us with information on the demographics of Bellevue College faculty and students. After obtaining demographic information, we researched what attributes result in a person being more at risk from COVID-19. Our research showed that the risk from COVID-19 is greater to those who are older than 60 years old, those who are male, and those who belong to certain racial minority groups (Native Hawaiians, Pacific Islanders, Asians, and Hispanics). After learning which groups are at higher risk from COVIDD-19, we determined the number of Bellevue College staff and students that fall into these groups. We found that there are around 300-500 total staff and students over the age of 60, and around 4,000 staff and students who are members of higher risk racial minority groups. As a result of our findings, we recommend that Bellevue College take special care to ensure the protection of those who belong to higher risk groups.

Task 3

The goal for this task was to “identify how healthy the students and campus employees are, compared to the general population.” To determine the health of the general population, we researched public data on the prevalence of underlying health conditions (asthma, diabetes, heart problems, stroke, chronic respiratory diseases, etc..). To determine the health of the Bellevue College faculty and students we created a survey that asked students to provide information about themselves, which included if they had underlying health conditions. From our survey, we were able to discern that 7.69% of Bellevue College students had underlying health conditions. This proportion is better than that of the City of Bellevue's population at 22.8%, King County's populations at 29%, and Washington state's population at 35.8%. Despite our findings, we were concerned that the survey population may not be representative of the overall Bellevue College population, as the number of survey respondents was small and only students participated. Also, the percentage for the surrounding areas was calculated by adding together the percentages of the independent health conditions listed on the King County Public Health Department and does not account for cases when a person has multiple conditions. Based on the analysis of our research and the survey, we concluded that Bellevue College's population is in a better state of health than the general population.

Despite our findings that Bellevue College's population is most likely healthier than the general population, we recommend that Bellevue College devise a more comprehensive method to determine the health of the Bellevue College community. We believe that obtaining more data would help college officials have a better understanding regarding the health of the BC community, and a more accurate perspective in comparison with the surrounding areas.

Task 4

Our goal for task 4 was to "find out whether schools have a higher transmission rate than general workplaces and community transmission in general." To determine the transmission rate of COVID-19 in schools, workplaces, and communities, we researched articles and scientific studies. Additionally, as we found there to be a lack of information on the transmission rate of COVID-19 in schools, we also researched studies and articles that detail the transmission rate of other illnesses in schools. Our research informed us that because COVID-19 is a relatively new infection, there have not been any studies or articles yet, which detail its transmission rate in schools. Nevertheless, we were able to collect some information from schools around the world that reopened under COVID-19 conditions. We found that some of the schools that reopened had little negative impact as a result. Despite that, those schools were attended by primarily young children, who are known to be less affected by COVID-19, and, therefore, the same conclusion should not be implied for higher education settings. Moreover, our research suggests that COVID-19 transmission could be greater in higher education due to young adults' tendency to be less concerned with health issues. Overall, we concluded that even though some schools have reopened with minimal consequences, some information suggests that schools may have a transmission rate higher than workplaces in general. Thus, we recommend that Bellevue College continue offering classes in an online or hybrid capacity as long as possible.

Task 5

Our goal for this task was to "determine what will happen to classroom instruction if professors get sick and how easily replaceable the professors are in this particular situation." This was an issue because the absence of professors would greatly affect the school and students depending on the amount of time the professor would be gone. With all of these concerns, our group decided to dive deeper into what a full-time instructor would look like at Bellevue college. We found out that full-time instructors have medical insurance and leave, they must have leave days to take time off when they become sick. However, if the number of professors who are ill rises too quickly, this

system would quickly get overwhelmed and it would be difficult to get classes covered. In conclusion, Bellevue College professors are not easily replaceable, and due to budget cuts, the amount of potential professors on staff is limited. Putting these professors at risk can impact staff, students, and education.

Task 6

The goal for task 6 was to “determine how comfortable students and faculty are coming back to live classrooms with PPE and without PPE.” We started drafting a short survey and also collaborated with professor Peterson to send it out to students and faculty members across the campus. The survey consisted of questions towards students and faculty related to their health and about returning to schools with and without PPE. We got 52 responses to the survey and as a team, analyzed the data. We found that a majority of the respondents were “not at risk of COVID-19”. Given that social distance guidelines are followed on campus, “39 respondents are somewhat comfortable, comfortable or not worried at all about attending on-campus classes”, while the remaining 13 respondents are either “uncomfortable going to campus or are not going to attend on-campus classes at all”. In addition, we also found that “48 people (92.31%) are willing to help sanitize their workplaces before and after their use and only 4 people (7.69%) are not willing to do so”. Based on the data we received from our survey, we have concluded that the majority of the people who responded are comfortable with attending on-campus classes, wearing a mask, and taking precautions to reduce infections. Also, we observed that a majority of the respondents do not have a medical condition that could cause them to be at a higher risk of catching the disease.

Task 7

The goal for this task was to “find out how many students are capable of and comfortable with online learning, how many need live class instruction to learn best, and how many students would prefer to continue remote instruction.” To figure out the answer to our question we included corresponding questions in the survey that we sent out to students and faculty. We then analyzed the data obtained from 52 students and faculty members. Our survey shows that out of 52 respondents, “39 respondents are either somewhat comfortable, comfortable or not worried at all about attending on-campus classes” while the remaining 7 are “are uncomfortable going to the campus or are not going to attend on-campus classes at all”. Also “26 respondents (50%) out of the 52 total, feel comfortable coming back to a live class, and 13 respondents (25%) feel uncomfortable”, while “the rest of the respondents (25%) hold opinions in the middle”. In conclusion, only a quarter of the students and staff who participated in our survey answered that they would not come to on-campus learning or would be

uncomfortable doing so. This shows that a majority of the students and staff are comfortable attending on-campus classes.

Task 8

For this task, our goal was to determine the technology that can help the campus reduce class sizes, so it can maintain social distancing. Brainstorm any other computer-based programs or ideas that could allow the college to reduce the number of students in class and improve online instruction. Due to the global pandemic, all schools were shut down and all in-person classes have been transferred online. To find out what computer software Bellevue College could use to help reduce class sizes and maintain a six-feet social distance, we began by researching software that offered video conferencing functionality. We found some programs that offered this feature. Bellevue College is mainly using “Zoom” for online instruction. However there have been some concerns with security issues, so we thought it was a good idea to inform Bellevue College. We also looked at the situation when instructors were sick. This would mean the instructor wouldn't be able to show up for a live lecture. We came up with the solution of using a webcam recorder to film the lecture beforehand. Software, such as Fillmore Scrn, Windows Camera, and QuickTime, are more commonly used ones. Thanks to the advanced technology, there is already an array of well-developed computer programs that are available for online teaching before the outbreak began. In conclusion, after taking a look at the different software, such as Google Meet, discord, Webex, etc. and listed features that each provides, we agreed that Google Meet was the ideal option due to its stability and security.

Task 9

For this task, our goal was to “determine what equipment and programs we would need to live stream classes to students at home”. This was an issue because we know that there are different requirements for streaming media effectively. With this knowledge, our group decided to look for websites that specifically mentioned what online learning required. Our group conducted research and came up with results that consist of purchasing cameras, PC or Macs, microphones as well as other hardware that might be beneficial for people who need to stream or hold their classes using the internet. During our research, we also see that providers such as Comcast are a great option when needing to use free internet. The conclusion our group came up with is that Bellevue College still needs to purchase many of the equipment mentioned above. Also if a student is in need of free internet then the company Comcast is offering that option currently. We recommend that Bellevue College purchase some of the equipment mentioned in our research.

Task 10

For this task, our goal was to “determine if the campus needs to upgrade and/or increase its servers and supporting software”. This was a problem that we needed to research because we did not know what the capacity of the online servers of Bellevue College. The burden of many people streaming their classes online could possibly bring BC servers down. The complexity of this problem did not allow us to do much research. We needed some assistance, meanwhile, we did research-based on what we know. Based on the best information available, Bellevue College uses 85 physical servers located on-premise and 173 virtual servers in the cloud. The on-premise servers most likely lack auto-scaling functionality, but because most servers are in the cloud, this does not seem to be a problem, as most cloud providers offer auto-scale services that are easy to implement. Due to lack of enough information, we cannot currently make a conclusion about the server capabilities, but it may be necessary for BC to move all its servers in the cloud, as the on-premise servers could be more costly to maintain and could fail due to increased traffic and the lack of the autoscale functionality.

Task 11

For this task, our goal was to “determine when (and if) faster internet technologies such as 5G or 4G may come online in the US, which could help with online instruction possibilities. Determine whether faster internet speeds are available or may become available till fall.” This is a concerning issue that we worry about because due to the COVID-19 closures people have to stay at home and they might have slower internet at home. Faster wifi and data coverage will help the internet reach more people for better use. Our research consists of finding out the difference in speed, performance, and stability between 4G and 5G. We found that 4G is available across Washington state, therefore students and staff can benefit from it. We also see that 5G is indeed faster than 4G as it reduces latency while making speeds faster. In conclusion, there is no clear timeline for when 5G will begin to deploy in the Seattle region. We recommend Bellevue College to stick with what they have while helping students reach internet availability.

Task 12

For this task, our goal was to “ determine what is the current method of tracking Coronavirus cases among Bellevue College community and research new ways of facilitating the task if needed.” For this task we contacted our team manager Professor Peterson who was able to contact the Bellevue College COVID - 19 response team, their research and knowledge were beneficial for our task as well. During our own

research, we found out that Bellevue College's responsibilities to track COVID-19 are mainly dictated by Washington State health official guidelines. Thus, Bellevue College's plan to track COVID-19 cases is composed of "manual contact tracing, mandatory health questionnaires, mandatory temperature readings, and self-reporting." As part of our research, we also discovered information on digital contact tracing, which is a technology that could potentially help Bellevue College track COVID-19 with greater accuracy and efficiency. Digital contact tracing is similar to manual contact tracing, where the spread of infections is tracked by the interactions between people. The difference with digital contact tracing is simply that the tracing is done by computers that can also use interaction data to track infections. The best digital contact tracing program currently is an API that Google and Apple have developed together. Though digital contact tracing would benefit Bellevue College, the technology is controversial. Thus, we recommend that Bellevue College continue to follow governmental guidelines, monitor digital contact tracing development, and ask government officials to adopt digital contact tracing.

Conclusion:

Overall, our research regarding various aspects of COVID-19 suggests that the virus presents a substantial obstacle to the continuation of normal Bellevue College programs. From our research, we have determined that the threat to faculty and students from COVID-19 makes it unlikely that Bellevue College can continue normal operations.

Introduction

To determine the best course of action for efficient education of students in the following quarter, Dr. Robert Viens, Assistant Vice President for Academic and Student Affairs at Bellevue College, asked us to research the available information on the COVID-19, assess the risks that the college would take if it were to reopen, analyze the alternatives to live to learn and present our findings and recommendations.

Currently, all classes at BC are taught online and there is no certain plan for reopening the campus. This situation is not ideal, as online education may not work for all students and faculty members perfectly and it may be more efficient to reopen the campus with certain safety measures.

Therefore, Dr. Robert Viens asked us to research how Bellevue College can safely re-open live classrooms fall quarter, assuming Governor Inslee authorizes school re-openings by that time. Specifically, Dr. Viens asked us to investigate seven questions:

1. What are the health risks for campus employees and students, based on the best, currently available information?
2. What are the demographics of the students and campus employees?
3. How healthy are the students and campus employees, compared to the general population?
4. What will happen to classroom instruction if professors get sick? How easily replaceable are the professors?
5. How comfortable are students and faculty coming back to live classrooms with PPE (personal protective equipment and other infection reductions measures we need to define through research) and without PPE?
6. How many students are capable of and comfortable with online learning? How many need live class instruction to learn best? How many students would prefer to continue remote instruction?
7. Do schools have a higher transmission rate than general workplaces and community transmission in general?

We found that the information about the virus evolves rapidly and a complete return to normalcy will not be possible until a vaccine is developed and the majority of the population has become immune to the virus, which scientists say

will not be available at least until early next year. Bellevue College has some high-risk groups, such as people older than 60 and members of certain racial minority groups (specifically, Native Hawaiian, Pacific Islanders, Asians, and Hispanics). Moreover, roughly 10% of Bellevue College's community has an underlying health condition and is at an increased risk of contracting a severe or life-threatening case of COVID-19. Additionally, one technology that could facilitate the efforts of BC to track infections in its community is digital tracing. However, this technology is controversial due to privacy and legality concerns.

Although some schools opened with little impact, we are not sure what kind of consequences reopening the campus would have on the transmission rate, as higher education schools have students who are older and have higher chances of being affected by the virus. Moreover, if professors were to get sick, it would not be easy to replace them and the instruction would be negatively impacted.

For online education, we agreed that Google Meet would be an ideal option due to its stability and security. The college would have to purchase cameras or microphone systems as well as capture cards, video encoders, and possibly new laptops as well. The vast majority of houses are connected to the internet, which is stable enough for basic online education.

We recommend the campus follows state directives and health officials' recommendations regarding social distancing measures. Bellevue College should keep online classes or hybrid classes format for as long as possible, instead of going back to the regular classes. This approach would ensure the safety of professors, older students, and about 10% of the population with additional health risks.

BC should start a program to successfully determine the general state of health of the students and staff by recurring a survey, for instance. Moreover, considering digital tracing could allow the campus to track existing cases of the virus.

To improve education quality and provide an efficient learning environment, it is recommended to try other computer software that offers different features and could help students learn more efficiently (Google Meet, for example). We will provide a list of items that have to be purchased to provide online instruction in our recommendation section. Meanwhile, BC should use its current internet

provider and focus on assisting students who do not have access to reliable internet.

Yet, if the college were to reopen, the campus should focus on the high-risk groups, mentioned above and provide students with protective equipment if necessary.

In the following sections, we provide additional details about our research methods, the results we obtained, the conclusions we drew from those results, and our recommendation.

Research Methods

We began our research by corresponding with our team's manager, professor Peterson, who helped us determine the scope of our research and identify the information the college needs to decide how to safely reopen. To ensure a comprehensive approach, we broke down our analysis into several tasks:

Task 1: Acquire an understanding of what the health risks for campus employees and students are, based on the best, currently available information.

Task 2: Obtain data on the demographics of the students and campus employees to better assess the health risks.

Task 3: Identify how healthy the students and campus employees are, compared to the general population.

Task 4: Find out whether schools have a higher transmission rate than general workplaces and community transmission in general.

Task 5: Determine what will happen to classroom instruction if professors get sick and how easily replaceable the professors are.

Task 6: How comfortable are students and faculty coming back to live classrooms with PPE (personal protective equipment and other infection reductions measures we need to define through research) and without PPE?

Task 7: How many students are capable of and comfortable with online learning? How many need live class instruction to learn best? How many students would prefer to continue remote instruction?

Task 8: Determine technology that can help the campus reduce class sizes, so it can maintain social distancing. Brainstorm any other computer-based programs or ideas that could allow the college to reduce the number of students in class and improve online instruction.

Task 9: Determine what equipment and programs we would need to live stream classes to students at home.

Task 10: Determine if the campus needs to upgrade and/or increase its servers and supporting software.

Task 11: Determine when (and if) faster internet technologies such as 5G or 4G may come online in the US, which could help with online instruction possibilities. Determine whether faster internet speeds are available or may become available till fall.

Task 12: Determine what is the current method of tracking Coronavirus cases among Bellevue College community and research new ways of facilitating the task if needed.

Below we give a more detailed explanation of how we conducted the research for each individual task and describe the reasoning behind each method.

Task 1: Acquire an understanding of what the health risks for campus employees and students are, based on the best, currently available information.

To evaluate the challenges that the coronavirus outbreak would pose for Bellevue College, we first identified different aspects of the pandemic that the college should take into account when deciding whether to reopen the campus, such as the current number of cases in the US, the current transmission rate of the virus, the challenges posed by the virus' transmission method, the impact of COVID-19 on different demographic groups, experts' predictions for the number of infections in the summer, projections of future waves, policymakers plans and strategies of how to reopen the economy, the progress on the vaccine development, available drugs approved for treating the illness, and the efficacy of acquired immunity to the virus, the chance of reinfection, etc. We then studied articles related to each of the areas thoroughly, along with their related consequences, and combined all the information into a summary that included all the points of view.

As much about the COVID-19 virus is still unknown and the situation is rapidly evolving, our methods consisted mostly of researching the newest information available and reading various articles related to it. We analyzed information pertaining to the general situation to gain a broad view of the pandemic, and we also conducted a more narrowed research targeting the local area specifically, to gain a perspective of how it is doing in comparison with the rest of the world.

As we expected, due to the novelty of the virus, there are many mixed scientific opinions regarding various aspects of the virus, such as the transmission method of the virus, available treatments, immunity to the virus, etc. Opinions are mixed

not only among different experts but also among the same experts, as discoveries are made almost daily and scientists understand the virus better.

Task 2: Obtain data on the demographics of the students and campus employees to better assess the health risks.

Because the risk posed by the virus varies with the age, gender, and race of the patient, we had to obtain and analyze the demographics of the Bellevue College community.

To assess the health risks associated with the opening of the campus, we first acquired demographic data, such as age distribution, racial distribution, and gender distribution. We contacted our team's manager, Pr. Peterson, who provided us with several primary sources that we later analyzed to get a sense of a demographic situation at Bellevue College. Our analysis was originally separate for both students and employees, but our final report mostly included the demographic data for the total population.

Then, we made several projections for every piece of data to estimate the risks. Specifically, we used the information, collected in task 1, and applied its percentages to the demographics of our campus. Based on the results, we provided a summary of the risks and attached a recommendation.

Task 3: Identify how healthy the students and campus employees are, compared to the general population.

To gain a sense of the general health of Bellevue College community, and the related risk from the new COVID-19, we conducted secondary research to determine the health level of Bellevue city residents, where the majority of students and employees live. We looked at Bellevue health data in comparison with residents from surrounding areas, such as King County and Washington State. For this task, we relied mostly on data provided by the King County Public Health Department. For health data related to Bellevue College employees and students specifically, we sent surveys across the campus. We then compared and contrasted the data and statistics to determine the level of health of the Bellevue College community.

Task 4: Find out whether schools have a higher transmission rate than general workplaces and community transmission in general.

To acquire information related to this task, we first performed secondary research to find direct studies that would report whether the transmission rate of COVID-19 is higher in schools than in other places. But, as we expected, because the virus is relatively new, we were unable to find direct studies that compare the transmission rate in school versus general settings. We proceeded by researching the transmission rates of infectious diseases in general in schools and what factors contribute to an increased transmission rate. Specifically, we looked at a study that performed contact tracing simulations in schools to determine if disrupting certain school routines and applying restrictions in certain areas of the school could lower the infection rates of transmissible diseases. Also, we looked at recent studies that reported transmission rates of COVID-19 after schools have been reopened.
We also performed research to understand what factors influence the transmission of the COVID-19 virus and whether these factors are more present in school settings. We looked at the virus characteristics, such as the way COVID-19 transmits and examined whether school settings provided a more favorable medium for transmitting the virus. We also examined whether certain student behaviors increase the chances of infection.

Task 5: Determine what will happen to classroom instruction if professors get sick and how easily replaceable the professors are.

To discover how the absence of professors would affect the school during the pandemic, we spoke directly with our team's manager, Professor Peterson, who is also a member of the faculty senate and has good knowledge about Bellevue College policies.

Task 6: Determine how comfortable students and faculty are coming back to live classrooms with PPE (personal protective equipment and other infection reductions measures we would define through research) and without PPE.

As soon as our proposal was approved, we drafted a short survey and we collaborated with professor Peterson to send it out to students and faculty members across the campus. The survey asked students and faculty questions related to their health and their comfort level regarding returning to school with

and without wearing PPE to school. We ended up getting responses from 52 people and we will be using and analyzing that data to come to a conclusion.

Task 7: Find out how many students are capable of and comfortable with online learning, how many need live class instruction to learn best, and how many students would prefer to continue remote instruction.

To find out the answers for this task, we included corresponding questions in the survey that we sent out to students and faculty. We analyzed the data obtained from 52 students and faculty members and made a conclusion based on the results.

Task 8: Determine technology that can help the campus reduce class sizes, so it can maintain social distancing. Brainstorm any other computer-based programs or ideas that could allow the college to reduce the number of students in class and improve online instruction.

Due to COVID-19, all in-person classes had been transferred to online. To find out what computer software Bellevue College could use to help reduce class sizes and maintain a six-feet social distance, we began by researching software that offered video conferencing functionality. We found a multitude of programs that offered this feature, so we only included a few that we thought were popular and better suited for Bellevue College to use for online instructions.

All the software included in the report requires no monthly subscription except for ClickMeeting. Although ClickMeeting requires its users to pay around 40 dollars per month, we thought it was a worthwhile investment for Bellevue College because it provides the feature of conducting live polls and Q and A, which could help facilitate and aid students' learning process.

Bellevue College is mainly using ZOOM for online instructions. We believed it was appropriate to inform Bellevue college of reported security issues with Zoom. To help support our argument, we provided the names of the companies and governments that had banned the use of ZOOM.

We also looked at the situation when instructors were sick and came up with the solution of using a webcam recorder to film the lecture beforehand. Software, such as Fillmore Scrn, Windows Camera, and QuickTime, are more commonly used ones.

In order to conclude which software is best for online instructions, we made a comparison table to have a detailed look at each one of them. We compared

things like cost, the number of participants allowed, and devices that software supports to help us decide which software we should recommend to BC.

Task 9: Determine what equipment and programs we would need to live stream classes to students at home.

Since programs were already covered in the previous task, this task will be focusing on just the equipment needed to live stream classes for students at home. To determine what equipment is needed to live stream classes to students at home, we began by asking Karrin Peterson, our team's manager, what the college's current plan regarding the equipment needed, as well as what equipment the college already had. Our team's manager told us that not every professor has a camera or microphone system yet. Using that information, we prioritized looking and shopping for camera and microphone systems.

Because there are various uses for camera and microphone systems, we decided to look for websites that specifically listed equipment for online learning. In addition to camera and microphone systems, we also found other equipment that could be beneficial for online learning, as well as alternatives to what the college already had. For example, we found capture cards and video encoders, which are pretty much a necessity in online learning. Also, even though the college already has desktops/laptops for the students and staff, we researched what specs are needed for desktops/laptops used for online learning.

For the most part, most of the resources we used gave us specific items, along with their pricing. However, for the items that did not, we either tried finding articles that compared the prices of each product or we used sites such as Amazon or eBay to find prices.

Task 10: Determine if the campus needs to upgrade and/or increase its servers and supporting software.

To determine whether the campus can support the transfer of all classes online, we corresponded with the Bellevue College IT Department through our team's manager, Professor Peterson. We also conducted an online interview asking various questions related to the Bellevue college server's capacity.

To understand whether BC can accommodate all of its services online we needed information regarding the way BC currently maintains its servers and whether the servers are capable of autoscaling on a need basis.

Task 11: Determine when (and if) faster internet technologies such as 5G or 4G may come online in the US, which could help with online instruction possibilities. Determine whether faster internet speeds are available or may become available till fall.

To determine if and when faster internet technologies would arrive in the United States, we first looked into what the current coverage is for wired and wireless internet in the United States. We looked at the advantages of 5G compared to 4G. We then researched the status of 5G deployment in the United States. From there we were able to narrow down our search to the status of 5G deployment in Washington State. We researched the accessibility of the internet in King County and how many households currently have internet access.

Task 12: Determine what is the current method of tracking Coronavirus cases among Bellevue College community and research new ways of facilitating the task if needed.

To learn the current methods that Bellevue College plans to use to track COVID-19 cases in the community, we researched information available online and corresponded with our team manager Professor Peterson.
Professor Peterson was able to request information on Bellevue College's COVID-19 tracking methods by contacting Bellevue College's COVID-19 response team. By researching online information we were able to gather further information on the methods of COVID-19 tracking that Bellevue College intends to use along with the requirements for those methods that are imposed by the Washington state government.

To discover what available technology could help boost the ability of Bellevue College to track COVID-19 infections in the Bellevue College community, we consulted online news sources and company releases to find technology that would improve tracking.

Research Results

In this section, we present research results for each of the tasks:

Task 1: Acquire an understanding of what the health risks for campus employees and students are, based on the best, currently available information.

Current transmission rate

According to data provided by John Hopkins University, as of May 19, 2020, there have been 5 million cases of COVID-19 worldwide, out of which 300,000 resulted in death (Taylor, 2020).

Since the beginning of 2020, when the US first entered the battle with the new COVID-19, until May 25, 2020, the virus has infected 1,542,309 people and killed 87,472 people in the US (Worldometers, 2020). Despite social distancing measures, which has dramatically decreased the number of transmissions, states across the US have seen tremendous surges in the number of cases. The number almost doubled from 859,318 cases with 44,014 deaths since April 24th, when we submitted the proposal to do research (Lutton, 2020).

Although Washington is the first state to combat the COVID-19 virus, it appears that the state has been gradually falling towards the bottom of the list by the total number of infections. The smaller number of total infections compared to other states shows that the transmission speed in Washington has slowed and that Washington state's containment efforts are working well. According to data posted on the Worldometers website, there were 18, 779 cases of COVID-19 infections in Washington.

Data provided by King County show that, as of May 15, 2020, there have been 7,347 confirmed cases of COVID-19 in King County, with a rate of 330 cases per 100,000 residents. There have also been 1,312 cases of hospitalizations (17.8% of positive cases) and 516 (7.0%) deaths (King County Public Health Department).

Nevertheless, although Washington is doing better in comparison with some other US states, the latest data shows that the infections could be on the rise

again in Washington state as well. An analysis of data collected between March 29 and April 15 in King County, previously indicated that the transmission rate had fallen under the critical threshold and that the effective reproductive number, which represents how many new infections stem from one single case, was below 1.0 (Public Health Insider, 2020). However, experts warn that as of April 22, this number is no longer falling and that it is, in fact, rising. New data from King County indicates that the reproductive number is now between 0.47 and 1.32, no longer below 1.0. A similar trend has been observed in eastern Washington, where the new number falls between 0.61 and 1.39.

Projections

Weighing in on the situation, Dr. Fauci, director of the National Institute of Allergy and Infectious Diseases and top advisor in the Trump administration, has warned that re-opening the states prematurely could lead to a devastating resurgence of the virus (Woodward, 2020). He also believes that, due to the virus being so transmissible and so widespread, there is no chance for the virus to be eradicated in summer, despite the best efforts to contain it.

On the same note, recent projections by experts from the Institute for Health Metrics and Evaluations (IHME) at the Washington University, show a sobering new estimate of 134,475 deaths by Aug. 4 (Quinn & KOMO News, 2020). The experts from IHME had previously projected this number to peak at roughly 72,000 by the same date. The experts believe that this spike is due to the increased mobility and ease of social distancing measures in many states. Christopher Murray, IHME Director, states that the risk of infection is still present across the entire US and that scientists expect the epidemic to last through the summer.

Additionally, a group of US experts has identified three different scenarios for how future waves of infections might evolve in the next 18-24 months (Woodward, 2020). The first scenario involves many medium waves happening at regular intervals during the next 18 months. The second and the worst scenario, which Dr. Fauci believes is inevitable, would be the result of prematurely relaxed social distancing measures and involves a much greater wave than the one the country has seen already, happening in fall and winter. The second scenario would require lockdowns measures imposed again. The third scenario is the best scenario and involves a “slow burn” of new infections for the next 18-24 months. Although experts believe this scenario is unlikely, it

remains a possibility. No lockdown measures would be necessary in this scenario.

Below are the visual representations of the scenarios described above:

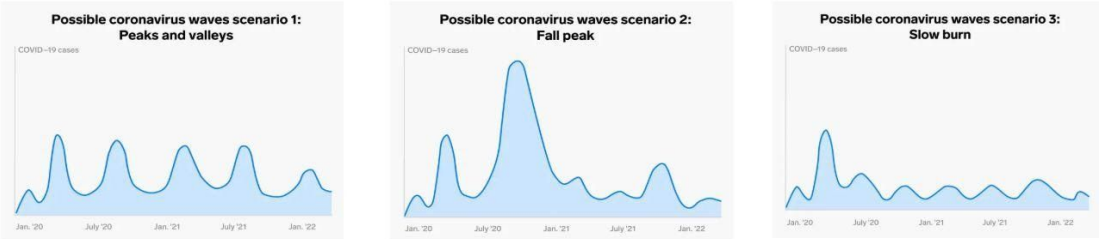


Figure 1 - Possible COVID-19 infection rate scenarios (source: Business Insider)

Ways COVID-19 transmits and related challenges

New York City's "COVID-19: Data" (2020) reports that the virus is transmitted from person to person, typically at a distance smaller than 6 feet. The virus travels through respiratory droplets that are emitted when an infected person coughs or sneezes, and lands on another person's nose, eyes, or mouth (CDC, 2020). Additionally, the virus can also be transmitted through contact with infected objects or surfaces when a person touches the surface and subsequently touches their face. Symptoms of infection include fever, cough, shortness of breath, and a possible loss of taste and smell, but infections can also be asymptomatic.

With that said, there has been no research-based evidence so far that describes how SARS-CoV-2 is exactly transmitted and much is still unknown about the virus (Brosseau, 2020). Many experts have voiced their frustration regarding the lack of scientific studies that determine the exact way the virus that causes COVID-19 transmits. This frustration is further fueled by some mixed messages and conflicting recommendations by the World Health Organization (WHO) and the US Centers for Disease Control and Prevention (CDC) about COVID-19 transmission routes.

Experts, such as Lisa Brosseau, professor of environmental and occupational health sciences in the UIC School of Public Health., state that a significant way of transmission for respiratory infections that are often overlooked is the close-range aerosol transmission (Brosseau, 2020). Aerosols are particles of different sizes, which contain viable infectious organisms, that are created as a result of talking, breathing, coughing, and sneezing. These particles can remain

suspended in the air for a prolonged period of time, enough for a person to inhale them and get infected. In addition, higher doses of infectious particles increase the chance of infection.

Brosseau says that large respiratory droplets (5 to 15 micrometers [μm]) can remain airborne for several minutes, while smaller particles (less than 5 μm) can remain in the air for hours due to them being less affected by gravity (Brosseau, 2020). Because large particles start losing moisture as soon as they are released in the air, they become smaller and more likely to be diffused. With time these particles are slowly dispersed throughout the space. Dr. Brosseau also states that the viability of viruses in the air dissipates with time, not distance, and depends on how robust a certain virus is.

A recent study found that aerosols from similar diseases, such as SARS-CoV, MERS-CoVSARS, can stay airborne for up to 3 hours, which is enough time for an infection to happen close to the source, as well as far from it (Brosseau, 2020). The expert outlines that short-range aerosol transmission by inhalation is probably more significant in transmitting respiratory infectious diseases than droplet transmission, contrary to what health organizations currently believe.

Dr. Brosseau advises that, while there is currently no evidence that supports a particular way of transmission over another, we should learn from previous experience with similar aerosol-transmissible diseases, and should focus on preventing short-range aerosol transmissions.

The mode in which COVID-19 spreads has raised health officials' concern about the pandemic. They believe that the way in which COVID-19 transmits is one of the epidemiologists' most dreaded scenarios (Sarah Kaplan, 2020). Two aspects of the virus transmission make the current situation challenging: its transmission route and its moderate severity. The virus's ability to transmit through the air makes it possible for new infections to grow more rapidly in comparison with other viruses which only spread through infected surfaces or blood. The ability of patients to remain mobile while infected greatly increases the chance of spreading the infection to multiple surroundings. The fact that cases can also remain asymptomatic in the beginning phase, or even for the whole duration of the illness, while still being infectious, remains another great challenge and complicates the efforts for containing the virus.

Another challenge that SARS-CoV-2 poses is that the virus replicates mostly in the upper respiratory tract in the first stages of the illness, as opposed to, other SARs diseases, which replicate in the lower respiratory tract (Brosseau, 2020). This makes the virus more efficient at transmitting through active shedding from sneezing and coughing. Later in the course of the illness, COVID-19 behaves more like SARS and replicates in the lower respiratory tract, thus becoming less infectious.

Furthermore, studies have revealed that the new virus spreads more easily than seasonal flu (WHO, 2020). While a person infected with the seasonal flu infects on average 1.3 other people, a person with the coronavirus infects between 2 and 2.5 new people, a metric known as the R0 value.

Demographics related risks

Data collected from mainland China shows that 4 out of 5 cases are mild and usually fully recover (Kottasová & Pettersson, 2020). Children, in particular, showed less severe symptoms than adults, with 90% of infected children being asymptomatic or showing mild symptoms. However, recent reports indicate a new trend in kids infected with COVID-19 (CDC, 2020). Multiple cases of Kawasaki inflammatory syndrome associated with Covid-19 have been reported worldwide. Health organizations, such as CDC, are currently investigating the incidents.

About 14% of the total cases developed severe complications, and 5% progressed to a critical condition.

The data also reveals that older people, typically those older than 60, are more at risk and that the chances of hospitalization and death correlate directly with age (Kottasová & Pettersson, 2020). However, new evidence emerging from France concludes that many hospitalizations involve patients younger than 60 years old.

Statistics from mainland China show the correlation between age and risk of mortality (Kottasová & Pettersson, 2020):

Age Range	Percentage (%)
0-9	0
10-19	0.2
20-29	0.2
30-39	0.2
40-49	0.4
50-59	1.3
60-69	3.6
70-79	8.0
80+	14.8

Table 1- Fatality rate by age

In addition, it seems that the virus is affecting men more severely compared to women (Kottasová & Pettersson, 2020). While the number of infections is almost the same for men and women, the mortality rate is higher for males, with men accounting for 63.8% of total deaths, compared to 36.2% of the total deaths for women.

Data provided by King County records on COVID-19, also indicates that there is a slight disparity in the way the virus affects different genders ("Daily COVID-19 outbreak summary", 2020). Using their demographic data, we estimated gender-related risks and summarized them in the following table:

Gender	Total number of people	Infected number of people (%)	Number of people hospitalized (%)	Number of Deaths (%)
Male	100,000	364 (.364)	71 (.071)	27 (.027)
Female		377 (0.377)	60 (.06)	24 (.024)

Table 2 – Estimated gender-related risks

(Note: Data is obtained on 6/5/2020 and may have dramatically changed since.)

Although the disparity appears not too significant, it seems to be consistent with data collected in China, and men seem to be affected more severely compared to women.

Those with underlying health conditions are especially at risk. The data collected from China indicates that over 1 in 10 people (10.5%) with cardiovascular disease die after contracting COVID-19 (Kottasová & Pettersson, 2020). Similarly, other diseases such as diabetes, chronic respiratory diseases, blood pressure, cardiovascular disease, etc., drastically increase the risk for mortality of people who have these diseases, compared to those without the disease. Data from Italy also shows that nearly all patients who died from COVID-19 had one or more underlying diseases, primarily blood pressure and diabetes.

The table below shows the chance of death for people with the above-mentioned underlying health conditions (Kottasová & Pettersson, 2020):

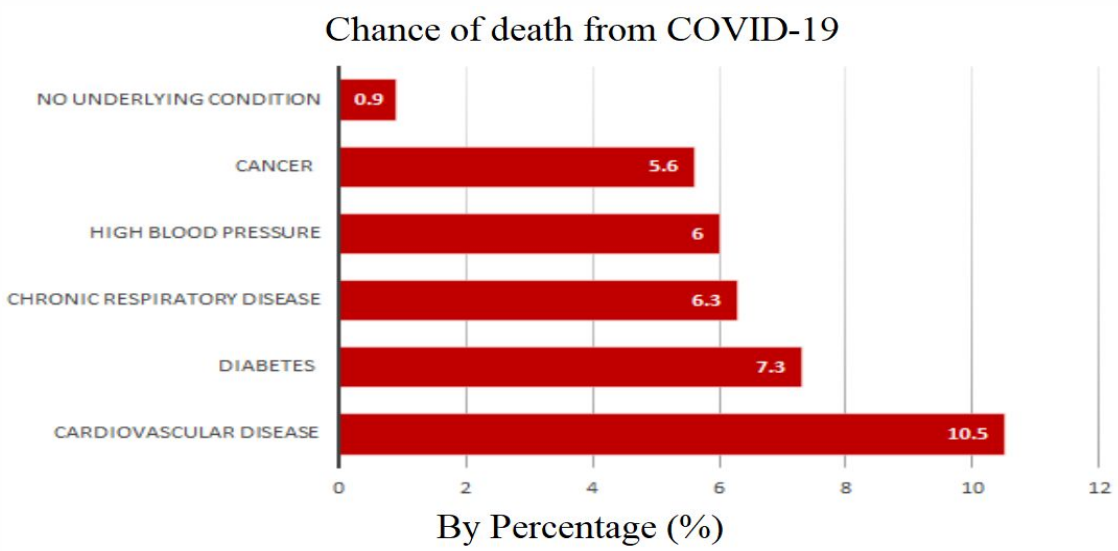


Figure 2 *- Chance of death from COVID-19 by percentage (%) (source: Chinese Center for Disease Control and Prevention)*

In addition, data provided by King County shows that COVID-19 is disproportionately impacting communities of color. Among the most affected communities are Native Hawaiian /Pacific Islanders, Hispanics, and blacks, while whites are among the least affected.

The data in the following table(see table 3) show this disproportion (the numbers are showing the cases per 100,000 residents and the dash (-) represents rates that are not statistically significant due to the small number of cases).

Community	Infections	Hospitalizations	Deaths
Native Hawaiian /Pacific Islanders	867.8	169.9	-
Hispanics	770.9	175.8	52.4
Blacks	419.4	74.4	-
Whites	166.2	40.3	20.8

Table 3- Infections, hospitalizations and death rates by community

Containment efforts

In order to slow down the rapid transmissions of the virus, many countries around the world have imposed social distancing measures, which, in turn, weigh heavily on the global economy. In the attempt to protect the economy, but at the same time fearing the second wave of infections if the economy is reopened, policymakers are devising a plan of how to safely resume economic activities with a minimal negative impact.

After modeling many scenarios, scientists concluded that alternating periods of strict lockdown measures with periods of more relaxed measures is one strategy that could work (Taylor, 2020). Fifty days of lockdown, followed by 30 days of eased measures appeared to be the most effective ratio. Even though this scenario would prolong the pandemic to more than 18 months, it would ensure that the death toll around the world would be significantly reduced. With that said, experts emphasize that countries should follow their own model that suits their needs best.

In his effort to gradually reopen Washington state and help its residents stay healthy, Governor Jay Inslee made an important step by allowing elective surgeries and dental services to resume, starting on May 15 (Drew). One condition was for providers to comply with rigorous health and safety protocols.

On the other hand, to prevent the number of infections from rising in Washington, Governor Inslee issued a new directive that went into effect on May 18 and which requires the general population to wear a face-covering in public, with some exceptions (Public Health Insider, 2020). This is an attempt at slowing down the transmissions by preventing infectious droplets from traveling from an infected person to others.

Additional efforts to contain the virus made by health officials include broadening the criteria for who needs to get tested (Public Health Insider, 2020). Public Health – Seattle & King County now recommends that anyone with mild symptoms should get tested, as opposed to previous recommendations to test only people at risk for severe illness, or healthcare providers, and first responders, due to supplies shortages.

In a recent analysis, health officials found that as of early May, 39% of infections occurred in people younger than 40, and 11% in people 19 and younger (Doughton, 2020). This is a significant shift from earlier reports which were stating that two-thirds of cases occurred in people older than 60. Health officials suspect that, among other causes, the new numbers are also the result of uneven testing during the course of the epidemic and that the new number reflects the increased testing capacity.

Immunity

Due to its novelty, many aspects of the virus are still a mystery. One aspect that scientists are desperately trying to find out is whether humans can acquire immunity to this virus and for how long. Acquiring this information would help scientists understand if the epidemic can be overcome by building herd immunity, a term used for when enough population has become immune to the virus and as a result, the transmissions rate falls naturally (Doughton, 2020).

With that said, experts warn that herd immunity is typically built through vaccination and that for the disease to stop spreading naturally, 70% of the population would have to be infected, a percentage that is highly dangerous as it would overwhelm the healthcare system and would increase the chance of death for those left without medical care (Doughton, 2020). The latest results based on a modeling show that there are currently less than 2% of people that have been infected so far in King County.

Efforts to determine whether humans develop immunity to the virus include testing for antibodies, as well as tracing cases of reinfection. None of these methods have yielded a definitive conclusion so far regarding the efficacy of our immune system when it comes to reinfection.

Antibody tests

An essential component in helping policymakers make decisions regarding when and how to reopen the economy are the antibodies tests. These would be helpful in answering critical questions such as whether a person already had the virus and whether they are safe to return to work without worrying about becoming sick (Alltucker, 2020). Although there was a lot of enthusiasm around the antibody tests, there has been some confusion created by the inaccuracy of tests developed by some companies. It appears that some labs failed to produce tests specific enough not to detect other viruses, and as a result, have achieved false positives and skewed the antibody results.

At this time, CDC is collaborating with other government agencies to assess the reliability of commercially manufactured antibody tests, while FDA has authorized emergency use of several antibody tests (CDC, 2020).

In addition, CDC informs that it’s unclear if those antibodies can provide protection (immunity) against getting infected again or for how long (CDC, 2020). They also inform that it is still uncertain whether antibodies make people immune to the virus and that it will take time to answer these questions. Also, while some people take up at least two weeks to develop antibodies, some people might not develop antibodies at all.

Because of this, the WHO has warned against issuing immunity passports based on the antibody tests (Alltucker, 2020). Even though the antibody tests would be of great help in fighting the pandemic, experts warn that making decisions based on these tests is only as reliable as the tests are accurate and that at this point it is still premature to rely on them.

Reinfections

Although rare, cases of reinfection have been identified. The US Navy has recently announced that 13 sailors from the USS Theodore Roosevelt ship retested positive after following a rigorous CDC guideline and testing negative for the virus twice before being cleared (McCammon, 2020). Similarly, Korean health

authorities identified cases of reinfection in about 163 patients, which makes up roughly 2 percent of the total recovered cases.

Health experts suspect that in some cases the virus could remain dormant in the human body and later reactivate. Another possible scenario is that tests are simply detecting dead virus particles that can no longer transmit or infect other people. But some patients who re-tested positive did see a relapse and became sick. They also believe that patients who get reinfected might have a milder form of the disease the second time and that their immune system has built at least some level of protection against the virus.

Vaccine progress

With the virus unleashed worldwide, there is little to no hope of bringing the number of infections to zero with the help of current measures, as they are not sustainable long term and weigh heavily on the economy. A return to complete normalcy and safety would require obtaining a permanent solution, such as the widespread availability of a vaccine.

Due to the worldwide spread of COVID-19, many scientists around the world are racing to secure a vaccine that could potentially be used as a preventive measure against the deadly virus (Radcliffe, 2020). Nevertheless, the process of making a vaccine is long and requires a vaccine to go through different phases, held at certain intervals apart, that test the vaccine for safety and efficacy before it is ready for use on the general population. This process usually takes up about 5 years. Despite that, there are currently many companies inside and outside the U.S. trying to expedite the process and about a dozen vaccine candidates which are currently going through the first phase of testing.

In the US, there are at least four or five vaccines that look encouraging, out of which the one developed by the NIH and Moderna Inc., as well as a different type created by Britain's Oxford University, are getting the most attention (Neergaard, 2020). In its first testing phase of Moderna’s vaccine, healthy, middle-aged participants developed antibodies similar to those of patients recovered from COVID-19, an immune response that scientists hoped for (Staff, 2020). These two seem ready to begin large-scale testing by July (Neergaard, 2020).

The Trump administration is also pitching in on the expediting effort through Operation Warp Speed, committing up to $1.2 billion and announcing its partnership with drugmaker AstraZeneca (Quinn, 2020). This will ensure 300

million doses or more of a COVID-19 vaccine developed by the University of Oxford. If the vaccine proves safe and effective, Operation Warp Speed plans to make available millions of doses of the vaccine by the end of 2020 or early in 2021.

Available treatments

While a vaccine is the permanent, preventive solution that scientists hope to obtain soon, they also search for other solutions to the virus which could help cure the disease and save lives in the meanwhile, such as different types of drugs (Radcliffe, 2020). Even though they also try to develop a drug that targets COVID-19 specifically, new drugs usually take up decades until they are approved for human use. For this reason, scientists are screening drugs that are already approved for other diseases as a potential treatment for COVID-19.

One type of drug that scientists have been researching is antivirals. This type of drug could cure people already infected by preventing the virus from replicating in the body (Radcliffe, 2020). There are currently several antivirals being tested for their effectiveness in treating COVID-19, out of which, Remdesivir has been approved for emergency use by the FDA, despite its mixed results, and can be used by doctors only. Additionally, a small company, Sorrento Therapeutics, is claiming that they also have a drug that has proven effective in blocking the virus and could be used in treating as well as preventing the disease.

In addition to antivirals, a few other types of treatments are being evaluated, such as drugs that trigger the immune system to attack the virus (Radcliffe, 2020). Among these types of treatments are monoclonal antibodies, a treatment that uses isolated antibodies from patients who survived SARS, and blood plasma transfers, a treatment that uses COVID-19 antibodies rich plasma from recovered patients to help the new infections. Stem cell use and immune suppressants are other treatments that are being evaluated. Other drugs that have been repurposed for use against COVID-19 and that have been approved for emergency use by the FDA, along with a warning against its use outside of medical facilities, are hydroxychloroquine and chloroquine. Opinions about the effect of this drug on the recovery process vary and have been mixed so far.

With that being said, all of these drugs are not a sure treatment for the virus and the risk for the general population is still great. Despite all the progress that has been made, distancing ourselves from others is currently still our only hope.

Task 2: Obtain data on the demographics of the students and campus employees to better assess the health risks.

Age-related risks

One of the factors that increase the risk of getting sick with the virus is age.

Figure 3 and Figure 4 represent the age distribution (Univstats, 2018) of students attending BC (see figure 3):

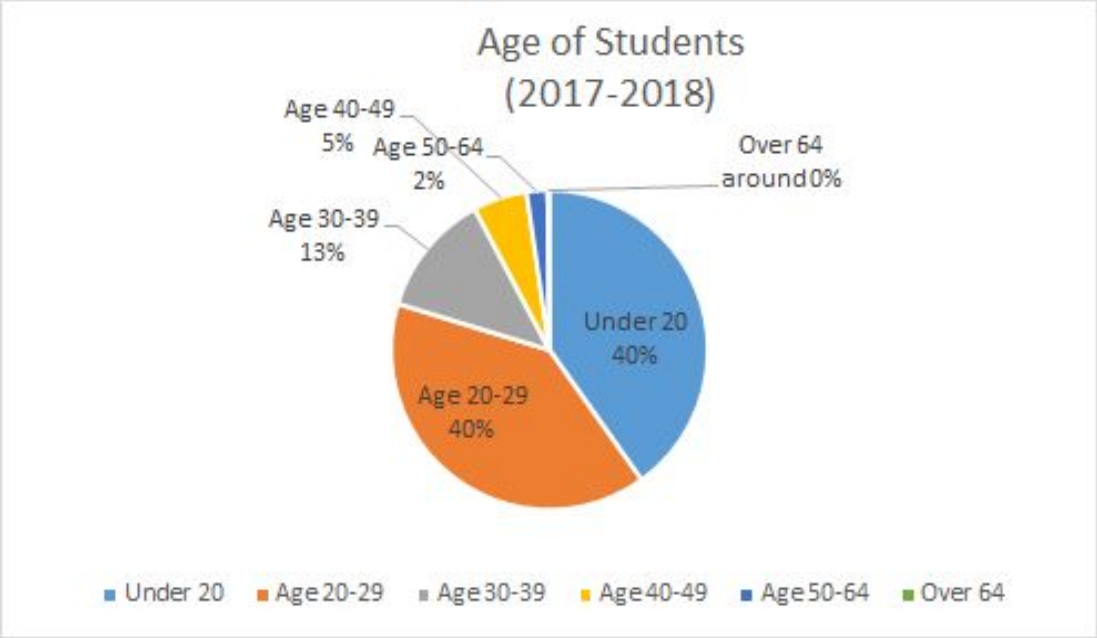


Figure 3 – Age Distribution of Students (Pie Chart)

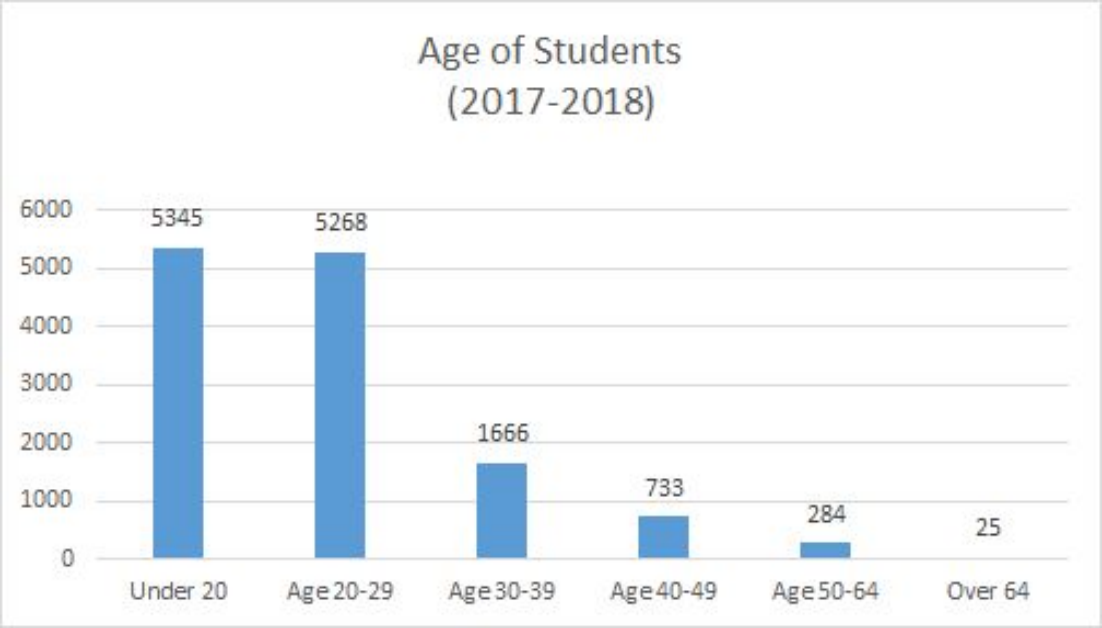


Figure 4 – Age Distribution of Students (Column Chart)

The students' age distribution falls mostly on the ages of lower risk. Students who are less than 30 years old make up about 80% of the total student population. Meanwhile, students who are most at risk only account for 2% of the campus' student population.

Employees' age distribution is shown in figures 5 and 6 below:

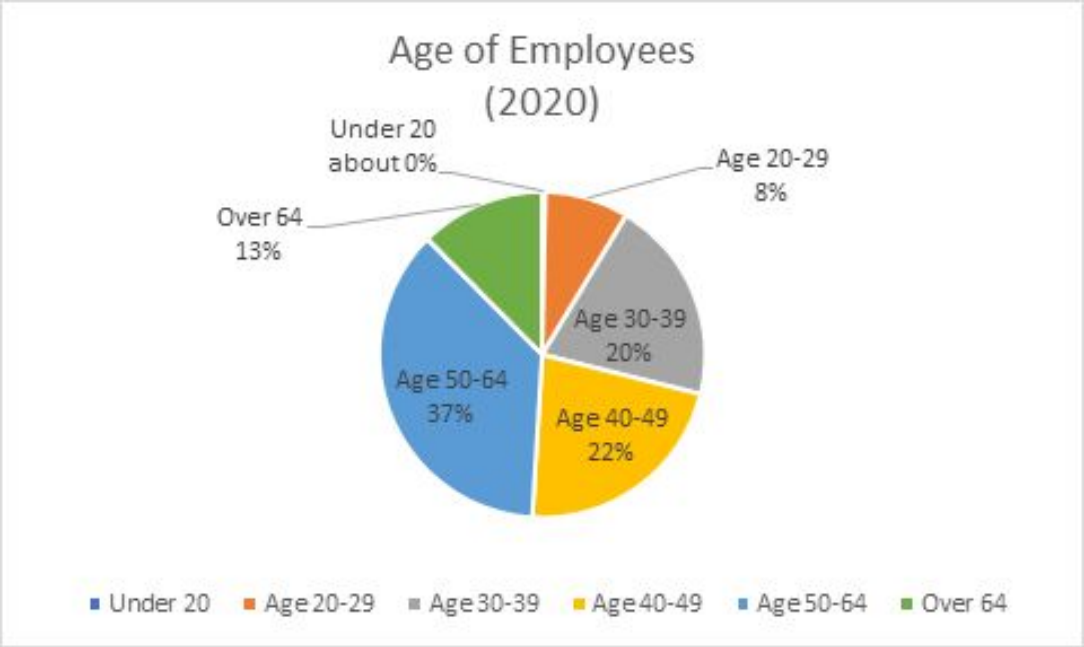


Figure 5 – Age Distribution of Employees (Pie Chart)

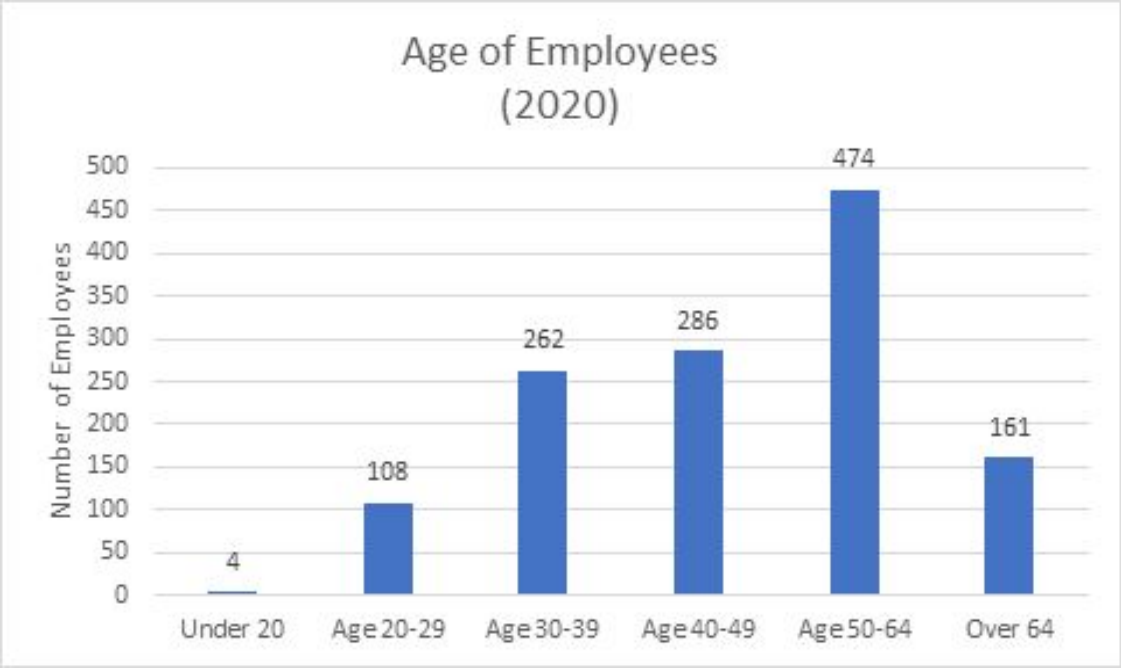


Figure 6 – Age Distribution of Employees (Column Chart)

A sizable portion of BC's employees falls in the group of high risk. About half of the campus's workforce is composed of people who are older than 50. The median age is 49.5 and the mean age is 50. There are 161 employees over the age of 64, making up 13% of the total number of employees.

Using the mortality rates for the people in the high-risk age range (over 60 years old), we generated a projection of how BC employees would be affected, if no protection measures are implemented. We present our findings in the following table(see table 4):

Age group	**Current mortality percentage among the age group (%)**	**Number of Employees at Bellevue College**	**Projected number of deaths at Bellevue College**
50-59	1.3	323	4
60-69	3.6	261	9
70-79	8	46	3

80+	14.8	5	0
Total	**-**	**635**	**16**

Table 4 – Estimated Mortality of Employees
(Note: *Younger groups of employees were not included in this dataset due to having statistically insignificant results (low percentage and low total numbers))*

If Campus opens without any protective measures, about 16 employees could be impacted fatally.

As concluded earlier, the age seems to be directly proportional to the higher risk of developing complications from the virus. Thus, while the majority of the student population is less likely to be severely impacted by COVID-19, the remaining 2% of the students and 50% of employees may require some protective measures.

Gender-related risks

The next factor of risk is the gender of a person. As mentioned before, men have a higher mortality and hospitalization rate when compared to women. Gender distribution of Bellevue College (Univstats, 2018 & BC, 2017) is shown below in figures 6 and 7:

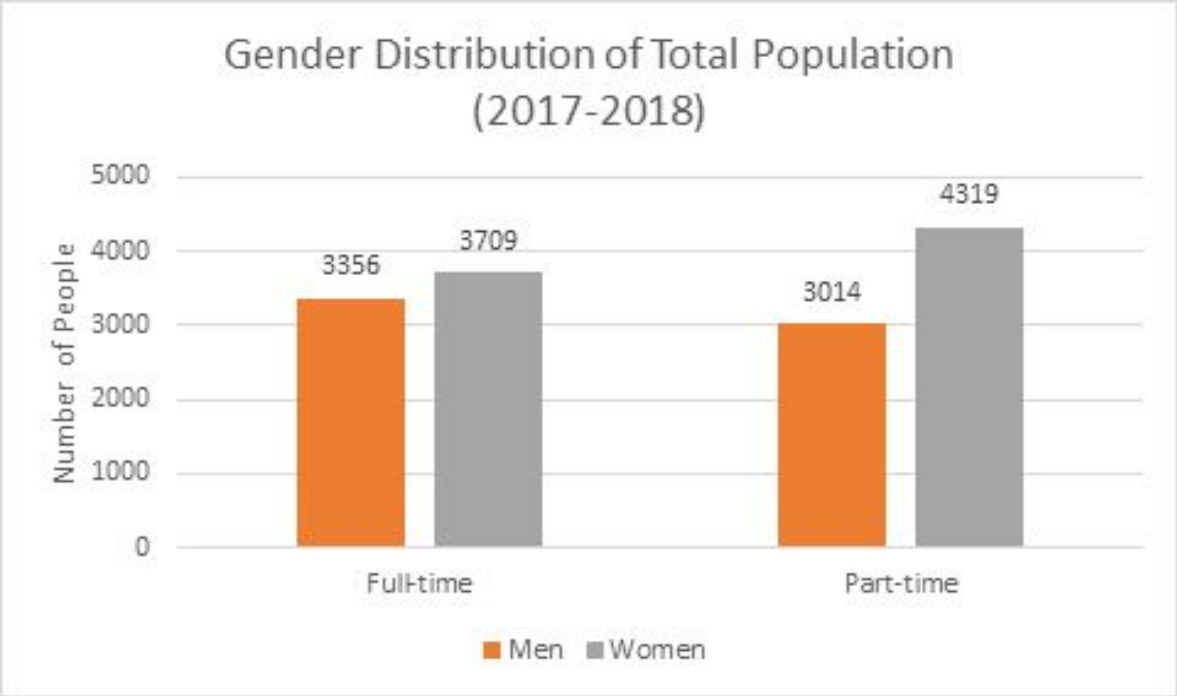


Figure 7 - Gender Distribution of Total Population (Pie Chart)

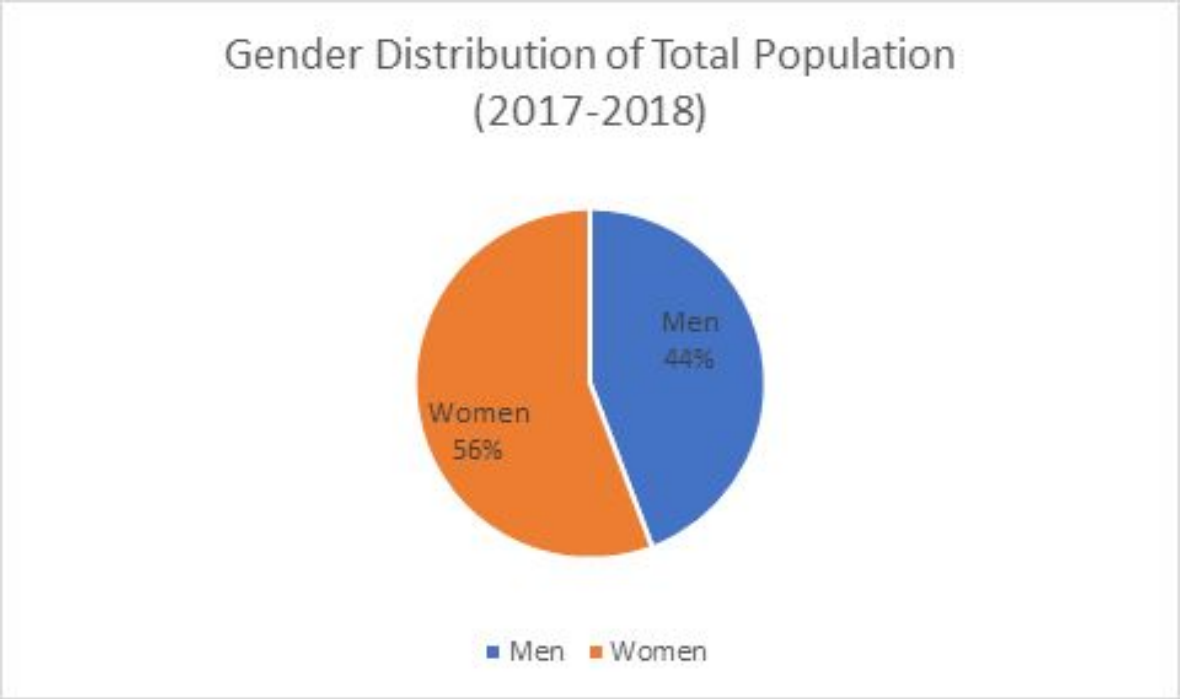


Figure 8 – Gender Distribution of Total Population (Column Chart)

Women represent the majority of the campus' population, accounting for 56% of the total number of people.

We applied the same percentage of infections, hospitalizations, and deaths rate as in King County for each group to gain a sense of an impact that Bellevue College might face due to the virus. Based on our projection, this is what the situation would look like:

Gender	**Total number of people**	**Infections** (Number of people)	**Hospitalizations** (Number of people)	**Deaths** (Number of people)
Male	6370	23	5	2
Female	8028	30	5	2
Total	**14398**	**53**	**10**	**4**

Table 5 - Estimated Risks for Total Student Population (Gender)
(Note: *Total number of people may be different due to primary data being from the year of 2017-2018*)

As a result of opening the college without any safety measures, a total of 53 people could be infected, 10 hospitalized, and 4 people could be fatally impacted.

Considering female prevalence on the campus, both populations would be affected almost equally, yet more women may get sick.

Race-related risks

According to the previously introduced data, White people will be less at risk of being impacted by the virus than other racial groups. Bellevue College's racial distribution is represented below in figures 8 and 9:

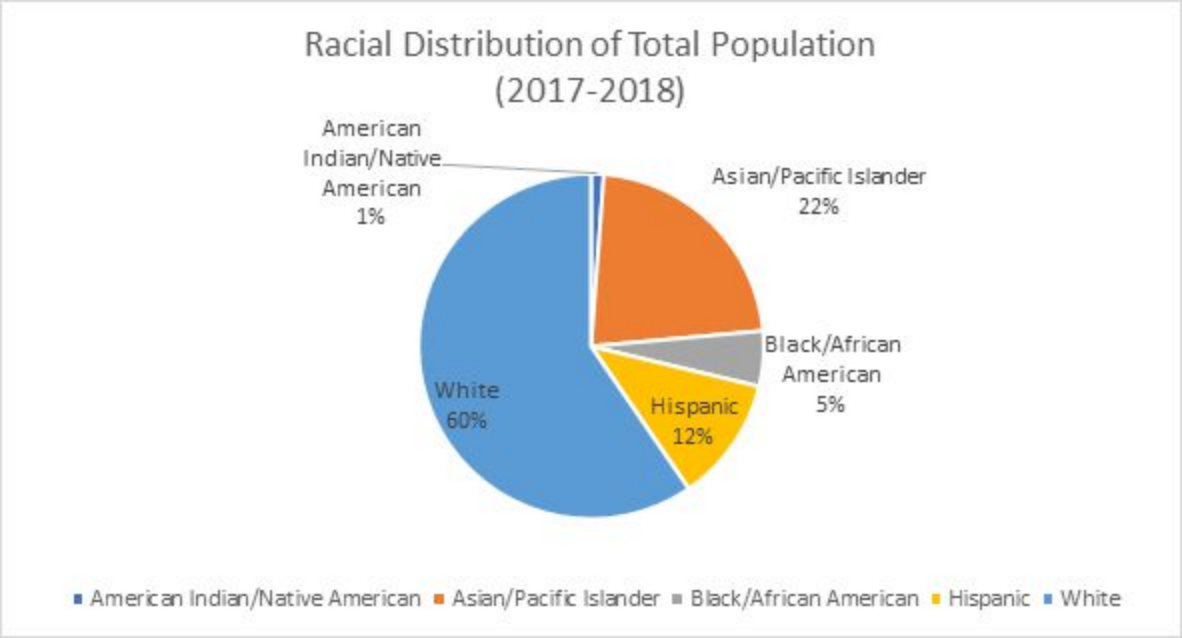


Figure 9 – Racial Distribution of Total Population at Bellevue College (Pie Chart)

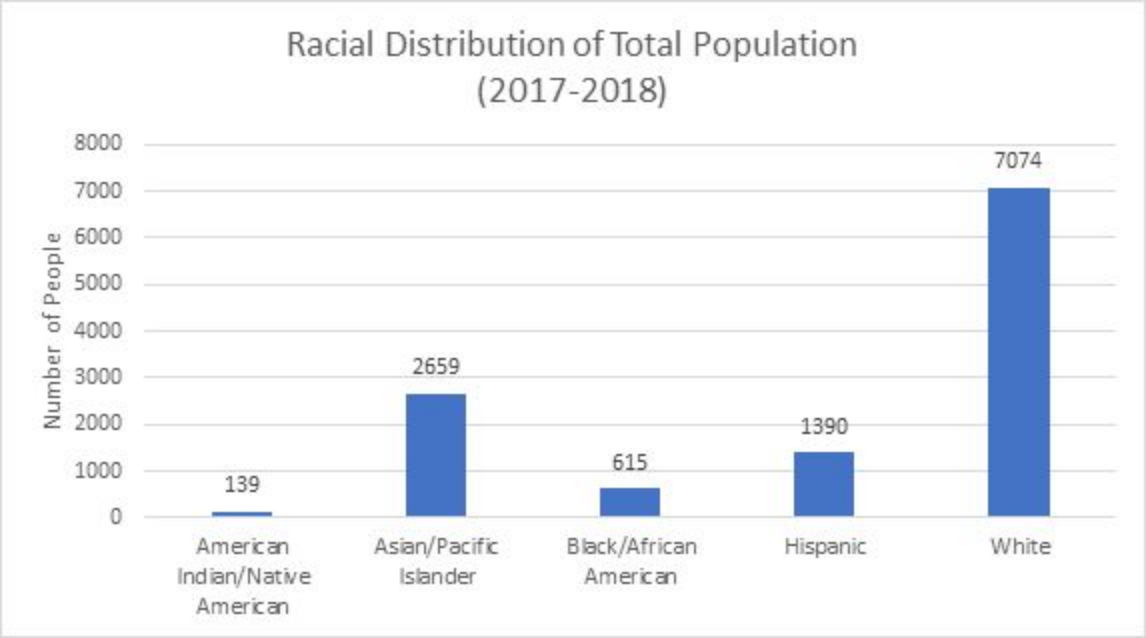


Figure 10 – Racial Distribution of Total Population at BC (Column Chart)

As visible on the charts, White people comprise most of the total campus' population. Following them would be Asian/Pacific Islander people (22%) and Hispanic people (12%), while African American people and Native Americans only make up 6% of the population.

Based on the current data collected from King County, these communities are disproportionately affected by the disease. There is a notable discrepancy in cases per 100,000 residents among these demographic groups. This is what the infections to hospitalizations and mortality ratios look like in King County(see figure 11):

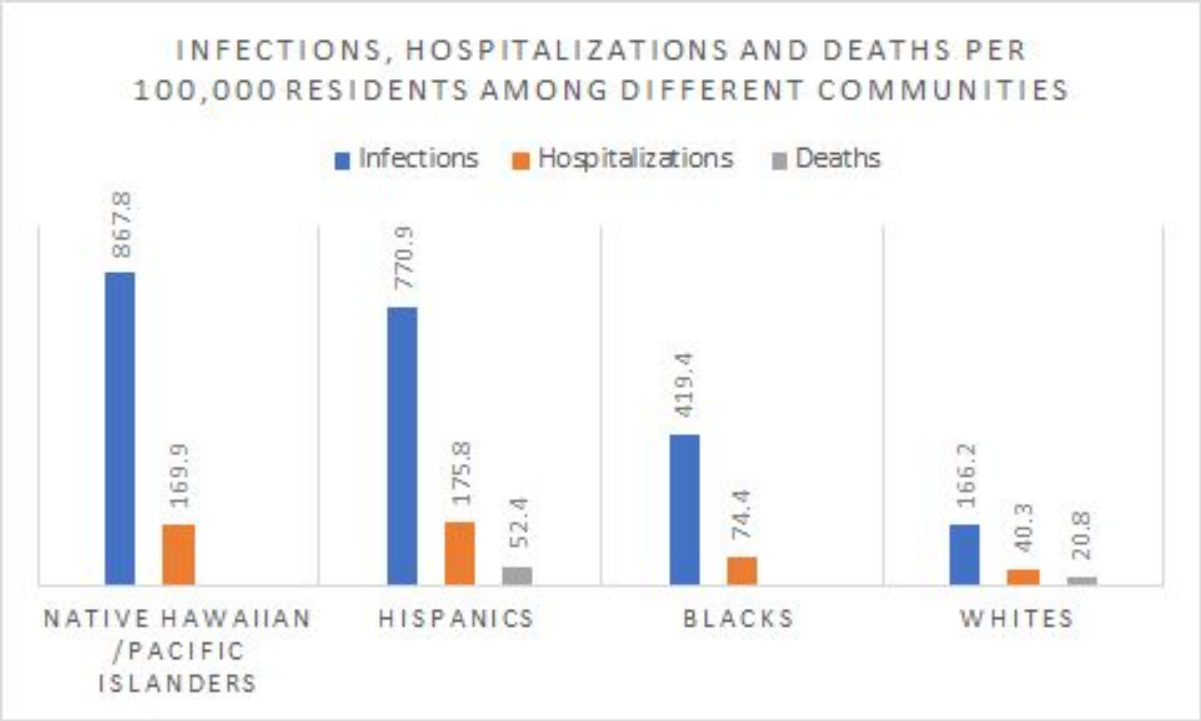


Figure 11 - Infections, Hospitalizations, and Deaths per 100,000 residents among different communities

We estimated the risks that various racial groups on our campus would face, based on the model from King County. We applied the same percentage of infections, hospitalizations, and deaths rate as in King County for each demographic group to gain a sense of how each demographic community would be affected. Based on our projection, this is what the situation would look like (see table 5):

Community	**Total** (number of people)	**Infections** (number of people)	**Hospitalizations** (number of people)	**Deaths** (number of people)
Native Hawaiian/Pacific Islanders/Asian	2659	23	5	-
Hispanics	1390	11	2	1
Blacks	615	3	0	-
Whites	7074	12	3	1
Total Population	**12928**	**49**	**10**	**2**

Table 6 – Estimated Risks for Total Student Population (Race)
(Note: *Due to the lack of data for Native Americans and its low population count at BC, they were not included in that estimation. Asian people and Native Hawaiian/Pacific Islanders were united into one category due to not being separated in the primary data.)*

The above estimation shows that there could be about 49 people infected, 10 of these people hospitalized, and 2 of these people could die as a result of opening the campus' without any safety measures.

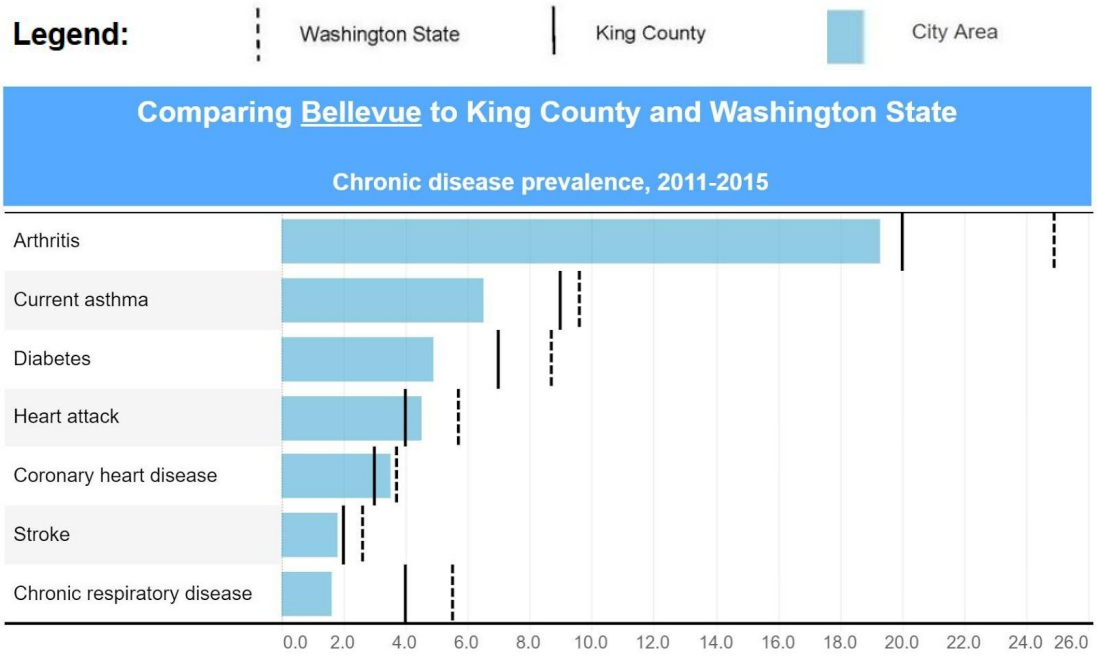
Task 3: Identify how healthy the students and campus employees are, compared to the general population.

Data gathered from the King County Department of Public Health shows that Bellevue city residents are generally healthier compared to the rest of the county and state. Chronic disease prevalence, such as asthma, diabetes, heart problems, stroke, chronic respiratory diseases, etc., is lower in Bellevue than in King County and the rest of Washington, although not necessarily significant (King County Public Health Department, 2019). For example, the rate of asthma in Bellevue is 6.5, much lower than 9.0 in King County and 9.6 in Washington state. The diabetes rate in Bellevue is 4.9, lower than 8.7 in Washington, and 7 in King County. Chronic respiratory diseases are also less prevalent in Bellevue, with a rate of 1.6, compared to 4.0 in King County and 5.5 in Washington State.

The lower prevalence is consistent for all the diseases when compared to Washington State. Compared to King County, the prevalence of chronic diseases is lower for many diseases, except for heart attack and coronary heart disease, which is higher in Bellevue than in King County. Heart attacks' rate is 4.5 in Bellevue compared to 4.0 in King County, and coronary heart disease is 3.5 in Bellevue, compared to 3.0 across all King County (King County Department of Health, 2020).

Adding the percentages for each chronic disease together, we can see that the total percentage of people with underlying health conditions is 22.8% in Bellevue, 29% in King County, and 35.8% in Washington.

The figure below, taken from the King County public department of health website, shows the prevalence of chronic diseases in Bellevue compared with King County and Washington state:



	Count	City Area Rate	King County	WA State
Current asthma	NA	6.5	9.0	9.6
Coronary heart disease	NA	3.5	3.0	3.7
Heart attack	NA	4.5	4.0	5.7
Stroke	NA	1.8	2.0	2.6
Diabetes	NA	4.9	7.0	8.7
Chronic respiratory disease	NA	1.6	4.0	5.5
Arthritis	NA	19.3	20.0	24.9

Figure 12 - Prevalence of chronic diseases in Bellevue compared to KC and WA (source: King County Department of Health)

Data showing the leading causes of death in Bellevue in 2011-2015 also indicates that the city ranks lower in the number of deaths caused by chronic diseases such as cancer, heart disease, stroke, diabetes, liver diseases, chronic lower respiratory diseases (King County Department of Health, 2015).

The figure below shows the leading causes of death in Bellevue compared with King County and Washington State (King County Department of Health, 2015).

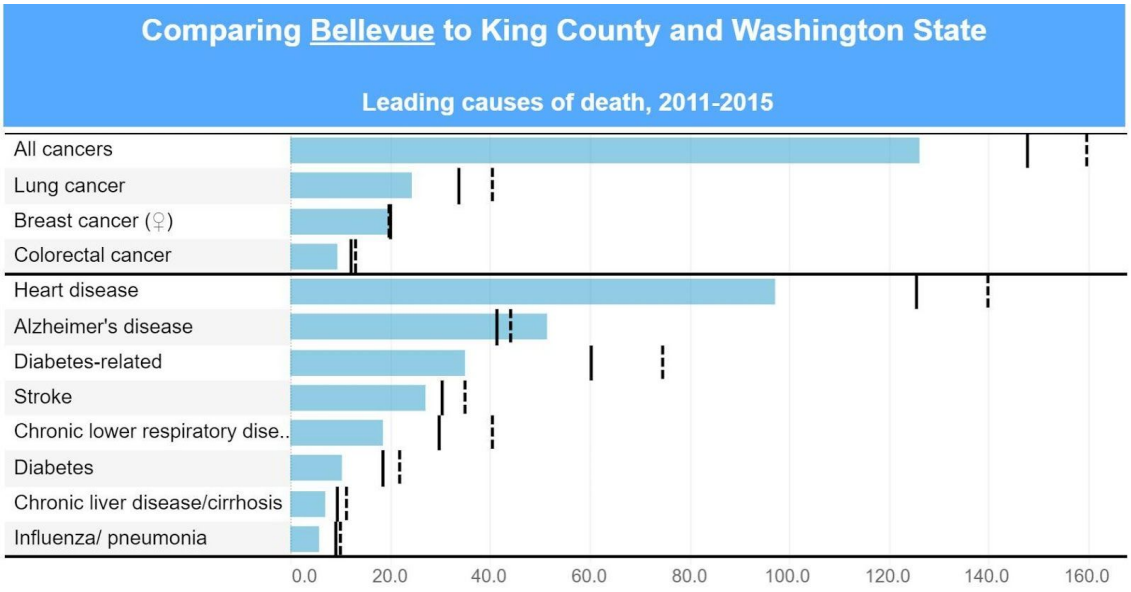


Figure 13 - Leading causes of death in Bellevue compared to KC and WA state (source: King County Department of Health)

Data obtained from the survey we sent across the campus shows that 7.69% of students had a medical condition that would put them at higher risk of developing life-threatening complications from COVID-19. More than 1 in 3 surveyees (34.62%) stated that they lived with someone with a medical condition that puts them at a higher risk of getting a severe or life-threatening case of COVID-19. A quarter of the surveyees also said that they visit someone with an underlying condition at least once a week.

When comparing the total percentage of people with underlying health conditions at BC, 7.69%, to 22.8% in Bellevue, 29% in King County, and 35.8% in Washington state, it appears that Bellevue College students and staff are in somewhat better health than the general assembly. But, with that said, the surveys were distributed to a small number of students on campus only and thus should not be relied on heavily. It also should be taken into account that many of

these students and employees have regular contact with someone with underlying conditions.

Task 4: Find out whether schools have a higher transmission rate than general workplaces and community transmission in general.

One of the first actions that many states took to stave the pandemic was to close down schools. Studies have shown that children play a central role in the transmission of the disease and can transmit the virus to each other and their families with considerable speed, as most cases in children, are mild or asymptomatic and are often overlooked.

While we did not find any specific studies that aim to directly find the transmission rate of Covid-19 in schools in comparison to general settings, we have looked at the transmission rate in similar settings. Recent studies found that households are the most commonplace for COVID-19 transmission (LaFaro, 2020). Data collected from Wuhan, China, also supports this theory. During the early stages of the Wuhan epidemic, most cases occurred mainly in public places and among family members living in the same household (EClinical Medicine, 2020). This is a potential indicator that school settings could increase exposure because school settings are similar to households in many ways.

The reasons why households and other indoor places have a higher rate of transmission are related to the way the virus transmits. According to the CDC, the virus spreads through proximity with an infected person, typically at a distance less than 6 feet. This means that crowded places, such as households and schools, where people sit at distances smaller than 6 feet from each other, are more likely to spread the disease. Because schools have a higher population density than other general places, with most regular classrooms being within an average of 24 students per 600sqft (a typical class size), there is little distance allowed between students. As a result, there is an increased chance of contracting the virus.
Emerging information indicates that aerosols could be an important factor in the transmission process (Brosseau, 2020). In places where the air is stagnant, airborne virus particles remain stuck inside, slowly dispersing throughout the space and with time reaching a wider area. Therefore, inside places that are surrounded by walls, such as classrooms, have an even higher chance of spreading the virus. In this scenario, even if students are sitting at a safe distance, the absence of air currents increases the chance of aerosols presence.

This could potentially expose students to the disease if a high enough dose of the virus particles is generated.

The figures below are a representation of the process described above:

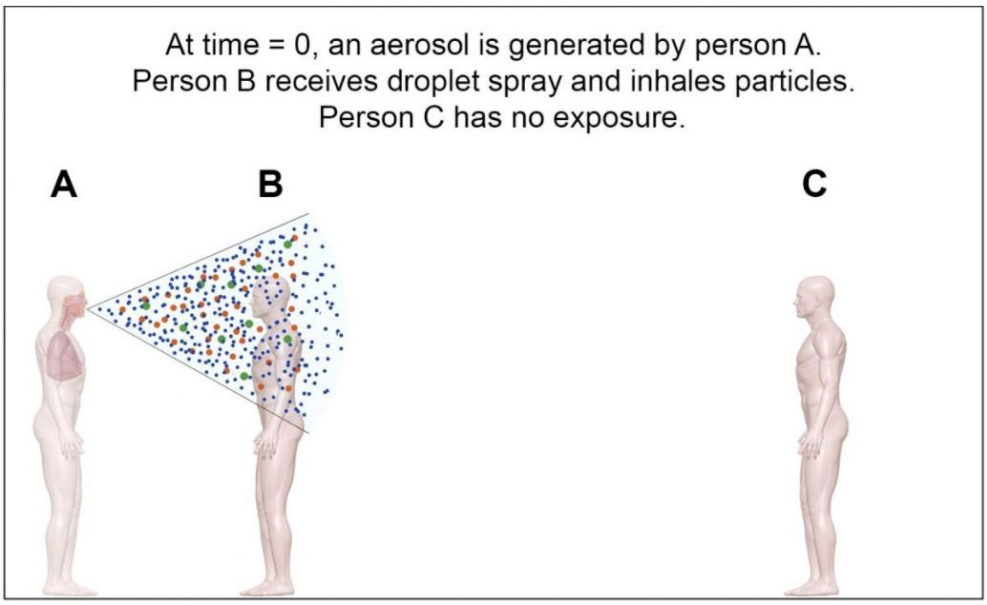


Figure 14 - Aerosols at time 0 (from the CIDRAP)

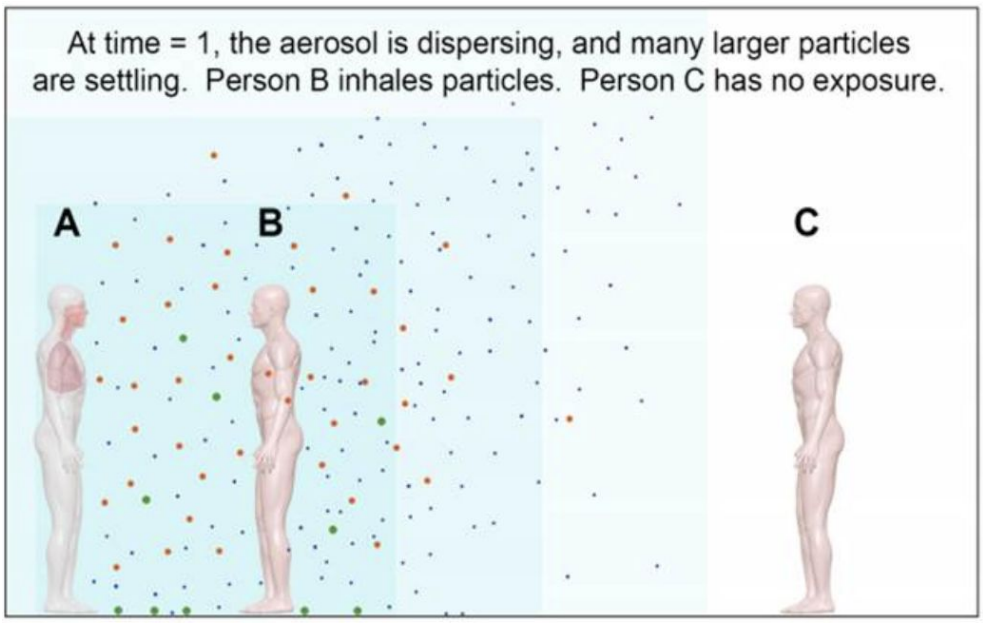


Figure 15 - Aerosols at time 1 (from the CIDRAP)

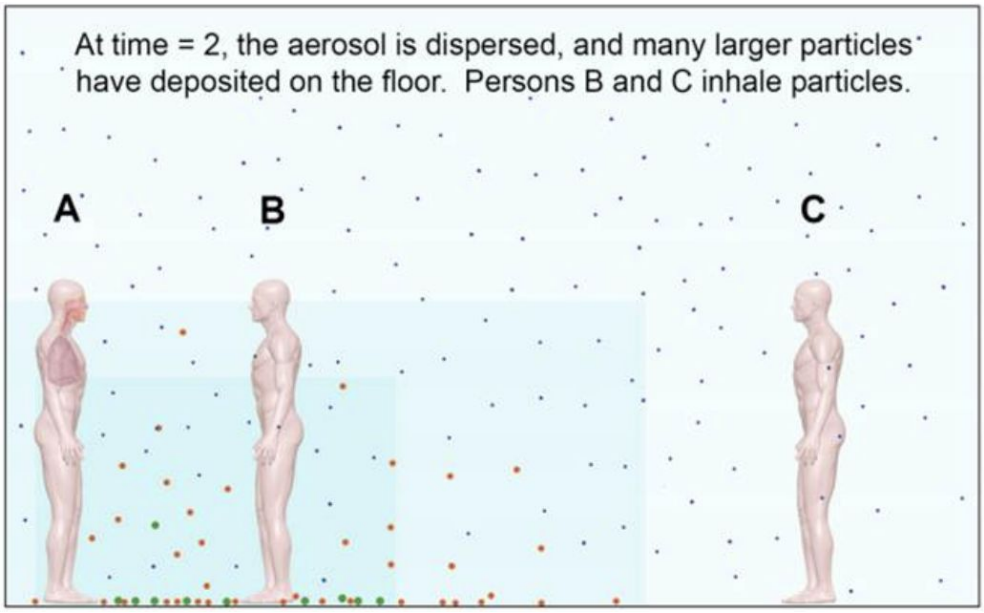


Figure 16 - Aerosols at time 2 (from the CIDRAP)

Another factor that could increase the transmission rate in college settings relates to the students' psychological and behavioral aspects. Young people tend to be less concerned about contracting the virus than older adults, whose risk of developing complications is higher (Doughton, 2020). Therefore they could be less concerned with respecting the distance and following adequate hygiene rules.

In addition to behavioral causes, some school routines can increase transmission rates in schools (Moreno, 2011). A study posted on the National Center for Biotechnology Information website, conducted in 2011, tried to determine the impact of certain school routines on the number of contacts each student has with others during the course of the day. The study was motivated by school closures in response to influenza but did not include the influenza transmission specifically in the calculations, and thus it applies to any transmissible disease. The experts simulated these routines without any restrictions at first to establish the baseline. Next, they simulated the same routines, but with some restrictions applied in various areas of the school, where the transmissions were more likely to happen. Below is a list of restrictions taken from the website (Moreno, 2011):

1. Baseline (B): The children can freely move about the entire school on their way to their scheduled rooms, sit anywhere they want in the classrooms,

and go wherever and with whomever they want in the schoolyard and the lunchroom. Every child has the same schedule.

2. Hall Restriction (HR): The children must stay in a defined walking area between their classrooms, the lunchroom, and the schoolyard (typically, the right-hand side of any hall as they travel). There are no restrictions while in the classroom, lunchroom, or schoolyard.
3. Classroom Restriction (CR): The children must remain seated while in their class. There are no hall, lunchroom, or schoolyard restrictions.
4. Schoolyard Restriction (SR): Each child stays in a randomly specified schoolyard area (with children from random classrooms). There are no hall, classroom, or lunchroom restrictions.
5. SR with Classmates (SRC): The children stay in a classroom-specific schoolyard area with their classmates. There are no hall, classroom, or lunchroom restrictions.
6. Lunchroom Restriction (LR): The children can eat only with their classmates. There are no hall, classroom, or schoolyard restrictions.
7. Different Schedules (DS): Each classroom follows one of three different schedules put forth by the school: the current schedule, a shift of 45 minutes, and a shift of 90 minutes. Other than that, the children can move throughout the entire school freely, sit anywhere they want in the classrooms, and go wherever and with whomever, they want in the schoolyard and lunchroom.
8. All interventions (AI): HR+CR+SRC+LR+DS.
9. Almost All Interventions (AAI): HR+CR+SR+DS.

The study concluded that when applying all these restrictions, the contacts among children was greatly diminished, as a result slowing down the spread of transmissible diseases. This study proves that some of the school routines are significantly increasing the transmission rate in schools and that applying restrictions in certain areas can serve as an alternative to school closures.

With that being said, during the time where schools were closed due to the COVID-19 pandemic, information has been gathered by some countries that carefully reopened schools, and new evidence has been found about the likelihood of children transmitting the virus. A report made by the Australian health officials concluded that children are unlikely to transmit the disease among each other, as well as to adults (Henriques-Gones, 2020). The report analyzed the transmission of COVID-19 in schools and childcare centers in New South Wales between March and mid-April. The report results are based on the

examination of 18 initial cases of COVID-19 infections (nine students and nine staff) from 15 schools. It found that no teacher or staff contracted the disease in these schools, and only one student in one primary school and one high school student may have contracted the virus. Even though 735 students and 128 staff were close contacts of the initial 18 cases, the results show a low transmission rate. The results could have also been impacted by many kids staying home by their parents. The results are also preliminary, though and need to be confirmed by peer review, but suggest that the transmission rate of Covid-19 within Australian schools has been very low.

Additionally, there have been other success stories, such as France, where schools have reopened without any significant rise in the number of confirmed infections (Boffey & Willsher, 2020). France reopened about 40,000 primary and nursery schools in mid-May, and in the first week, only 70 cases of COVID-19 had been reported.

Despite these studies, most of the world remains in lockdown. As some countries start to ease their restrictions, many start thinking about school and start asking themselves, "What does a classroom look like during a global pandemic?" Only places like China and Denmark began reopening their schools. Face coverings are a common sight as schools reopen in those parts of the world. Students are sporting face masks with few exceptions, like during the gym class and lunch. Another solution is to move the class outdoors. In Denmark’s case, they started reopening schools in mid-April; classes are outside as much as possible such as in parks (Deliso, 2020). The World Health Organization (WHO) recommends daily temperature checks for students as part of schools' COVID-19 public health measures. In Shanghai, students and staff enter the school through thermal scanners. At Montana's rural Willow Creek School, one of the first schools to reopen in the U.S., students' temperatures are checked upon entrance (Deliso, 2020).

With that being said, because children are known to be less affected by COVID-19, it would not be recommended to draw the same conclusions for college campuses, as students in these settings belong to different age groups. Older students could exhibit a different reaction to the virus.

By and large, there is a lot of controversy going about what will happen in fall when schools typically start a new year. An expert task force focusing exclusively on school closure is trying to help inform states and counties that are struggling to make this enormously consequential and urgent decision regarding the fall

semester (Christakis, 2020). They should review the state of the evidence regarding horizontal transmission among children and their families, as well as what is known about the feasibility of distance learning and the psychological implications of children continuing to stay at home.

Task 5: Determine what will happen to classroom instruction if professors get sick and how easily replaceable the professors are.

To discover how the absence of professors would affect the school during the pandemic we spoke directly with our team's manager, Professor Peterson. According to Professor Peterson, full-time instructors make up about 35% of the active professors during a normal quarter. The other 65% are considered adjuncts, these professors do not have leave time or extended absences through the college, this could put further strain on the continuity of instruction (Peterson, 2020). This means that 65% of instructors will not be paid if they get sick, this leaves the majority of instructors at a greater risk compared to professors who are considered full time.

In the case of full-time instructors, although they have medical insurance and leave, they must have leave days to take time off when they become sick. This leaves even the fully covered professors at potential risk. There is the possibility of a professor donating sick leave time to a fellow professor but this is not a dependable scenario. Thus, it is unclear what would happen in case a professor, who is covered but has no sick days left to take, would get sick. In the case that an instructor does become ill, the department chair will request that another professor in the same department substitute during that period (Peterson, 2020). However, if the number of teachers who are ill rises too quickly, this system would quickly get overwhelmed and it would be difficult to get classes covered.

It is important to note that professors at Bellevue College cannot find substitutes easily as there is no pool of substitute teachers readily available, and the college cannot hire substitute teachers, as opposed to the public school systems (Peterson, 2020). Moreover, for a professor to substitute a different class they must possess a graduate degree as well as specialization to teach the curriculum. According to Professor Peterson, substitutes must have a degree in the curriculum that they are covering. This limits the support for teachers even more as departments are limited to the number of professors they have staffed.

Due to the pandemic, Bellevue College has initiated budget cuts as well (Peterson, 2020). These budget cuts will affect all departments at the school causing further strain on professors. These budget cuts come with a hiring freeze which could prevent some departments from hiring instructors who could potentially act as a substitute. Overall, as the pandemic gets worse so could Progress report for Dr. Viens May 31, 2020 Page 42 conditions for professors and the continuity of instruction for students.

The pandemic poses a serious risk to the Bellevue College professors, not only is the majority of the staff paid less and have no sick leave, but they also are not provided health insurance through the state. Ultimately, if too many professors fall ill it may cause them financial hardship as well as causing several classes to be canceled for weeks or longer depending on the severity of the illness. This could seriously impact both the friends and family of the professors as well as the learning and education received by Bellevue College students.

Task 6: Determine how comfortable students and faculty are coming back to live classrooms with PPE (personal protective equipment and other infection reductions measures we would define through research) and without PPE.

For the most part, the majority of the respondents are not in any situation where they might be at risk of getting COVID-19. When asked if they had a medical condition that made them have a higher risk of getting COVID-19, 48 people (92.31%) said no. 34 people (65.38%) said that they did not live with someone that had a medical condition that made them have a higher risk of getting COVID-19. There were also 39 people (75%) that stated that they did not regularly visit someone that had a medical condition that made them have a higher risk of getting COVID-19.

Given that social distancing guidelines are followed, 6 people (11.54%) are uncomfortable going to campus and 7 people (13.46%) are not going to attend on-campus classes at all. On the other hand, the remaining 39 people are either somewhat comfortable, comfortable, or not worried at all about attending on-campus classes. To be more specific, 13 people (25%) are somewhat comfortable, 15 people (28.85%) are comfortable, and 11 people (21.15%) are not worried at all about attending on-campus classes.

As for how comfortable with how people are wearing masks, only 2 people (3.85%) are unwilling to wear masks, whereas the rest of the people are willing to wear masks, all of which have mixed feelings about doing so. 7 people (13.46%) think it will impact their learning, 9 people (17.31%) would prefer not to wear

them, 11 people (21.15%) are willing to wear them and go to places where people do not wear them, and 23 people (44.23%) believe they must absolutely be worn.

Regarding safety precautions to ensure that the virus does not get spread around, 48 people (92.31%) are willing to help sanitize their workplaces before and after their use and only 4 people (7.69%) are not willing to do so.

Task 7: Determine whether students prefer online or on-campus learning.

Our survey shows that out of the 52 total respondents, six people (11.54%) are uncomfortable going to campus, and seven people (13.46%) are not going to attend on-campus classes at all. On the other hand, the remaining 39 respondents are either somewhat comfortable, comfortable, or not worried at all about attending on-campus classes. To be more specific, 13 respondents (25%) are somewhat comfortable, 15 respondents (28.85%) are comfortable, and 11 respondents (21.15%) are not worried at all about attending on-campus classes.

In summary, 26 respondents (50%) out of the 52 total, feel comfortable coming back to a live class, and 13 respondents (25%) feel uncomfortable. The rest of the respondents (25%) hold opinions in the middle. It seems the results are related to the rate of online class preference. Two respondents (3.85%) feel they can't motivate themselves to work in an online class, and nine respondents (17.31%) say they can sort of use online classes, but don't believe they are receiving the same quality of education. The rest of the respondents (26.92%) out of 25 people who don't have a preference for online classes, feel that online classes are OK, but they do a bit better in live classes. In summary, the twenty-five people that make up 48.08% of the survey don't have a preference.

Task 8: Determine technology that can help the campus reduce class sizes, so it can maintain social distancing. Brainstorm any other computer-based programs or ideas that could allow the college to reduce the number of students in class and improve online instruction.

Due to the coronavirus outbreak, universities across the U.S. made the unprecedented decision to close their campuses and to transfer instructions online. Schools' faculty members and staff are trying their best to adjust themselves to this sudden and unexpected change. Thanks to the advanced technology, there is already an array of well-developed computer programs that are available for online teaching before the outbreak began.

Based on information obtained from our team's manager, professor Peterson,

Bellevue College is currently using the following online meeting programs: Zoom, MS Meetings, and ConexED, to accommodate online learning. We did research to find out if any other programs might be a better alternative.

We managed to gather information about currently used software as well as new software in order to gain a perspective of which software is the best option. We took a look at the following programs:

Google Meet

Google Meet, previously known as Google Hangouts, is a video-chatting app specifically designed for professional use, enabling people to chat over video and text. Before the coronavirus pandemic, it was already popular among businesses and organizations. Google Meet allows up to 250 participants to be on the same call and participants can share their screen and turn their camera and audio on and off anytime they want. Faculty can also invite their students to the video conference they hold by adding each student's email.

Discord

Discord is an online chatting platform that allows its users to communicate through voice and text, and it can be accessed through phone, tablet, and computer. It does require you to download the software if you are operating it on your phone or tablet. First-time users are expected to create an account before beginning using the service they provide on their platform, but everything is free of charge. Discord is an ideal place for online learning because it lets its users create multiple channels for different purposes in one server, which makes organizing class material becomes an easy task. Its voice channels enable users in the same server to instantly voice chat with each other, which is great for group discussions and assignment help.

Webex

Webex is a web conferencing app that allows people to virtually meet with each other. It requires no download and can accommodate up to 100 people in one conference call. Most importantly, it has recently been made free due to the coronavirus pandemic. Its easy-to-navigate interface will help instructors make the transition of moving live class instruction to online learning smoothly.

Skype

Skype is a software designed by Microsoft where individuals and businesses can make high-resolution group video/voice calls, send messages, and share documents. One of its notable features is that it can generate subtitles as the person is speaking. This feature can help facilitate the learning process and enable students to concentrate more on what the speaker is communicating.

Facebook Messenger

There are billions of active Facebook users around the world, which explains why Facebook messenger is one of the top choices for online instructions. It requires no subscription and there is no time limit on video calls. Most importantly, Facebook has ensured Rooms' privacy and protection, with the ability to monitor who sees your room, and you can lock and unlock so that no intruder can enter. That way, faculty would not need to worry about the class being interrupted and students can learn better.

Panopto

Panopto is a video capture software developed at Carnegie Mellon University. 22 out of 25 top U.S. universities are currently using this program to record their lectures and upload those reordered lectures to canvas, blackboard, and moodles. One of its unique features is search within videos, which allows students to search for words that are spoken in videos and fast forward to that part without having to watch the entire video.

Blackboard Collaborate

Blackboard Collaborate is an online synchronous interactive platform that can provide users with a fully interactive Web conference environment and use private and public chat, whiteboard, application sharing, and addition and editing anytime and anywhere through your computer, tablet, or mobile device.

ClickMeeting

ClickMeeting is the only software program here that requires a monthly subscription fee. Depending on the number of participants one is planning to have in one video call, the cost can go up as high as 100 dollars per month. Holding a class with 25 – 30 students should only cost about 40 dollars per month. Although its service is not free of charge, it is worth investing in

considering the features that it offers. Aside from the common features of screen sharing, group messaging, and call recording, it also allows its users to conduct live polls and Q & A (question and answer) sessions to make the call more engaging.

ZOOM's Security Concerns

Due to the COVID-19 outbreak, the number of daily users of ZOOM has skyrocketed from 10 million in December to 200 million in March. As more and more users are relying on ZOOM for online meetings, security issues begin to emerge. There are numerous cases of ZOOM bombing, strangers intruding online meetings, and data leak. Security issues like these have led some companies and governments to ban the use of ZOOM. Therefore, Bellevue College should consider using other alternative software to create a better and safer learning environment for its students.

Companies that no longer use ZOOM

- Google
- SpaceX
- Smart Communications

Governments/Government agencies that no longer use ZOOM

- Taiwan
- German Foreign Ministry
- United States Senate
- Australian Defense Force

In the table below we summarize the information and give a detailed comparison of the software, outlining the main feature and aspects, such as cost, maximum participants, screen sharing, call recording, group messaging, as well as what devices support the software, and other noteworthy features.

	ZOOM	MS Teams	ConexED	Google Meet	Discord	Webex
Cost	Free	Free	Free	Free	Free	Free
Maximum Participants	1000	250	100	100	At least 30	1000
Device Supported	Mobile & Computer	Mobile & Computer	Mobile & Computer	Mobile & Computer	Mobile & Computer	Mobile & Computer
Screen Sharing	√	√	√	√	√	√
Call Recording	√	√	√	√	√	√
Group Messaging	√	√	√	√	√	√
Other Special Features	React with emoji on screen	Direct access to email	Presentation streaming	Background noise filtering	Voice channel	N/A

	Skype	Facebook Messenger	Panopto	Blackboard	Click Meeting
Cost	Free	Free	Free	Free	$40/month
Max Participants	50	50	N/A	>100	30
Device Supported	Mobile & Computer	Mobile & Computer	Mobile & Computer	Mobile & Computer	Mobile & Computer
Screen Sharing	√	√	√	√	√
Call Recording	√	√	√	√	√
Group Messaging	√	√	√	√	√
Other Special Features	Auto-generated	N/A	Search inside videos	Virtual whiteboard	Q&A

	Subtitles				Live Polls

***Table 7* - Software comparison**

Each software has its advantages and disadvantages, but since Bellevue College is using them for educational purposes, Google Meet seems to be best suited for hosting online classes because it is more stable and secure than any other existing software program, provides multiple ways of accessing it without any download, and offers the unique feature of background noise filtering. The private information of students and instructors should be properly protected and since Google Meet was originally designed for companies involved in healthcare and finance, it has video and recording encryption that can effectively prevent anonymous users from entering calls without the permission of the host. Google Meet is extremely user-friendly. Users can easily access conference calls held on Google Meet from their Gmail account, google calendar, and google slides, which can help improve students' willingness to attend class meetings. Student's microphone accidentally picking up background noise is a common issue during online classes. It not only causes distractions but also disrupts students' concentration. With Google Meet's background noise filtering service, online instructions can be carried out smoothly without being interrupted.

In addition to programs that allow for synchronous instructions, we have also looked at programs that could help with instructions offered asynchronously, such as software that could be used to make videos and pre-record a class. Teachers can upload the pre-recorded lecture videos to canvas or YouTube when they are unable to show up to class. That way students can continue learning new material despite the absence of their instructors. Having lectures recorded and uploaded online can also help with reducing class size if attendance is not mandatory.

Webcam Recorder

The following pieces of software provide the service of recording videos at no charge, allowing instructors to film the lecture beforehand:

- Filmora Scrn
- Windows Camera
- QuickTime

In the table below (see table 8), we collected the basic information regarding the

cost of using them and the important features that they have.

	Filmora	Windows Camera	QuickTime
Cost	Free	Free	Free
Webcam Recording	√	√	√
Audio Recording	√	√	√
Video Editing	√	x	√

***Table 8* - Webcam Recorder Comparison**

Task 9: Determine what equipment and programs we would need to live stream classes to students at home.

The pandemic caused by the coronavirus outbreak has forced many schools across the United States to not only shut down but to adapt to online learning as well. Because of this, many schools are going to purchase hardware to make this transition to online learning easier for both the staff and students.

According to our team manager, Karrin Peterson, Bellevue College currently plans to use student computers on the receiving end and Dell computers with 8MB hard drives for each classroom. Due to the coronavirus outbreak, many internet services such as Comcast and Xfinity are offering their internet and WiFi services for free. As a matter of fact, Bellevue College's technology resources page has some of these services listed on it for student and staff use. However, not every professor has a camera or microphone system yet (Bellevue College 2020).

The following will consist of information about equipment that the college could use, as well as alternatives to what the college is currently using:

- Cameras (optional, not needed if there are phone cameras):
    - Why they are necessary (Barcelos 2018):
        - Although some laptops/desktops do have a webcam built-in, they may not be helpful for teachers who require hands-on demonstrations. For example, webcams may not fully

capture teachers who do work on whiteboards or demonstrate how lab equipment works. Cameras (or phone cameras) can show what the teacher wants the students to see more efficiently.

    - Some options are (Telestream 2015):
        - Canon Vixia HF R400 ($200)
        - Canon VIXIA HF R800 (under $300)
- Capture cards (optional, not needed if there are phone cameras or if cameras do not need them):
    - Why they are necessary (Cooper 2020):
        - A lot of cameras require capture cards. In fact, most do not even work without them. It is important to make sure if any cameras purchased require a capture card. Sometimes, the need for a camera card can be the deciding factor as to whether or not a camera should be purchased, especially if it is on a budget.
    - One option is (Telestream 2015):
        - Blackmagic Design Intensity Pro ($190)
- Microphones (Telestream 2015):
    - Why they are necessary:
        - Even though some cameras may come with built-in audio, the sound quality may not be as great. This can negatively affect the learning of students, especially those with disabilities.
    - Some options are:
        - Audio Technica ATR2100 ($60)
        - Pyle-Pro PDMIC58 ($13)
        - Shure PG185 Lapel Mic and Wireless System ($299)
- Video Encoder (optional, not needed if there is already a software video encoder):
    - Why they are necessary (Managuelod 2020):
        - Video encoders compress livestream videos as they are being recorded. This helps prevent buffering, lag, and an overall grainy video. This is most helpful for users with a slow Internet connection. Although hardware video encoders may cost more than software video encoders, hardware video encoders perform better.
    - Some options are (Brown 2020):
        - VidiU ($699)

        - LiveShell X ($699)
- Computers/laptops for both teachers and students (Telestream 2015):
    - Why they are necessary:
        - There are a number of students who relied on Bellevue College's on-campus computers and wireless services due to a lack of Internet and/or devices at home. Now that there is an outbreak, many students no longer have this access and will need it to do online learning. Even though Bellevue College does currently plan to provide both teachers and students with laptops and internet access, there does not seem to be a specific type of laptop chosen for the students. A lot of students will need specific specs for their laptops, some may even prefer one operating system over another. Students must be given what they are most comfortable with so they can adapt to this new learning environment easier.
    - Any Mac or PC with the following specs will be sufficient (at least $600):
        - High definition display: 1280 x 720
        - i5 Intel Processor
        - 8GB RAM
        - 1GB Video RAM

Task 10: Determine if the campus needs to upgrade and/or increase its servers and supporting software.

Due to BC's cancellation of live instruction in response to COVID-19, many classes and services are now being offered remotely, and many more students, faculty, and other employees are now accessing online services compared to regular times. Based on the best information available to us, Bellevue College currently has 173 virtual servers maintained in the cloud and 85 servers hosted on-premise. It is possible that many of the services used in online instruction have increased traffic to both types of servers. Because most of the servers are virtual and maintained by cloud providers, it is easy to increase their capacity on a need basis, as most providers offer easy auto-scale functionality. It is possible though that the servers hosted on premise are over-solicited as well and that the increase in traffic exceeds the servers' capacity. This could result in a failure to accommodate all the required services.

Task 11: Determine when (and if) faster internet technologies such as 5G or 4G may come online in the US, which could help with online instruction possibilities. Determine whether faster internet speeds are available or may become available till fall.

5G compared to 4G

To find out if and how Bellevue College would benefit from faster internet technologies, we first conducted research to understand what the difference between 5G and its predecessor 4G is. We also looked at the current coverage of 5G in the United States and researched if new coverage would be added by the start of the fall quarter at Bellevue College and in the surrounding areas.

We found that 4G technology is already available across Washington State, including the Bellevue College area. Students and faculty can take advantage of this technology with almost any smartphone available today. While 4G does allow for Internet access in even remote areas of the state, it is limited by its speed, capacity, and latency. While it is more than adequate for accessing the web and performing other standard actions, data-intensive activities such as viewing live streams can become unreliable if there is too much traffic on the network.

The increase in speed from 4G to 5G can be significant. It uses a new spectrum range known as the millimeter-wave. It can support a larger amount of traffic on the network and reduces latency (“Everything You Need to Know About 5G”, 2020).

5G Availability

5G technology is beginning to roll out across the United States. Cities such as San Francisco, New Orleans, Baltimore, and even Spokane have access to 5G from at least one provider, yet Seattle is notably absent from that list (Looper, 2020). The City of Seattle stated although the telecommunications industry has goals of bringing 5G to Seattle, there is no timeline available regarding when that will happen (“Frequently Asked Questions: 5G in Seattle”, 2019).

In the interest of accelerated 5G deployment, an order was issued by the Federal Communications Commission limiting the prices cities can charge 5G providers for attaching 5G technology to city utility poles. Bellevue agreed to the proposed limit and stated some providers can start rolling out 5G, while Seattle is not adhering to the order (Nickelsburg, 2020).

Internet Availability

According to the 2018 Technology Access Study run by Washington State, 95% of Seattle households can access the Internet using wired or wireless services ("Broadband in Washington", 2020). On average Seattle, households pay $141 per month for Internet access. Twenty-four percent of Seattle households report having accessed the Internet at the library and/or a school, college, or university within a month prior to the survey. With COVID-19, the majority of these locations are closed due to lockdown restrictions. While there is no specific measurement available, there is a possibility of overlap between those who cannot access the Internet at home and those who lost access to the Internet at schools and libraries due to COVID-19 closures.

Task 12: Determine what is the current method of tracking Coronavirus cases among Bellevue College community and research new ways of facilitating the task if needed.

After corresponding with our team manager Mrs. Peterson, and analyzing information published by Bellevue College, our team has discovered that Bellevue College will be required to track COVID-19 in the methods outlined in the Washington State Phase 1 Higher Education & Critical Infrastructure Workforce Training Restart COVID-19 Requirements.

The Phase 1 Higher Education & Critical Infrastructure Workforce Training Restart COVID-19 Requirements necessitate that Bellevue College creates a Phase 1 exposure plan ("Phase 1", 2020).

The Bellevue College public safety page outlines the school's plan as follows:

Before attending their class or shift on a given day, students and faculty will be asked to complete a mandatory questionnaire and undergo a temperature reading. The questionnaire will ask the individual to report their health condition and identify if they have symptoms of COVID-19 infection.

If a student experiences COVID-19 symptoms, they will be asked to report their symptoms to their instructor and seek medical attention. The student's instructor will then be asked to complete an infection report and send it to the CARE team. The CARE team will subsequently follow up with students. The CARE team will inform students and instructors of their contact with an infected individual and ask them to self-quarantine as per CDC guidelines.

Similarly, faculty who become infected will be required to report their infection to Bellevue College's HR department. The HR department will report infections to the lead COVID-19 supervisor and inform students and instructors of their

contact with an infected individual and ask them to self-quarantine following CDC recommendations.

Aside from current tracking methods, an important technology that could benefit Bellevue College in tracking COVID-19 is digital contact tracing. Digital contact tracing is a technology where people's electronic data is used to inform health officials of whom may have come into contact with infected individuals. Unlike manual contact tracing, digital contact tracing is not reliant on infected individuals' memory to report who they came in contact with. Instead, digital contact is much more concrete and allows for more accuracy.

Research we did into the use of digital contact tracing revealed the technology to be somewhat controversial (Servick, 2020). We also found that digital contact tracing programs face legality issues and as a result appear to be mainly designed for adoption by governing bodies. As such, we found that it may be unlikely that Bellevue College could directly adopt new programs and apps.

Currently, there are many digital contact tracing apps and programs either developed or in development that could benefit Bellevue College.

The most significant digital contact tracing program our research found is Apple and Google's joint COVID-19 tracking API. The API is a set of programs that act as a framework for which governments can build a digital contact tracing app ("Apple and Google", 2020). What is notable about Apple and Google's API is that it allows for data to be gathered from Android and Apple devices at the same time.

Google and Apple are also in the process of building a more robust digital contact tracing program into the software of their phones ("Apple and Google", 2020). The software would be opt-in and would use Bluetooth technology to map interactions between individuals. If a person becomes infected with COVID-19 they would enter it into the software and the app would notify everyone who has come into contact with the person over the past 14 days.

Conclusions

In this section, we present the conclusions of our research regarding each of the tasks:

Task 1: Understand the health risks posed by the virus to campus employees and students.

Due to its novelty, opinions regarding many aspects of the virus are mixed and vary not only among different experts but among the same experts, as more information about the virus emerges and scientists understand it better. Although experts are trying hard, it will be a while until they establish the exact nature of the virus, such as the exact methods it transmits, how it acts on our bodies, as well as until they find treatments that are effective in combating the virus, develop a vaccine that would build immunity against the virus, and determine how well the immune response can protect us against a second re-infection.

Although governments have imposed unprecedented social distancing measures and, as a result, the transmission of the virus has slowed tremendously, the virus has infected five million people worldwide, as of May 19, out of which 300,000 resulted in death. With a reduced infection rate, policymakers are looking to slowly ease the social distancing measures and allow for some normalcy to return. A complete return to normalcy will not be possible until a vaccine is developed and the majority of the population has become immune to the virus, which scientists say will not be ready at least until early next year.

With that said, the infections are still ongoing and the danger still persists.

Task 2: Obtain data on the demographics of the students and campus employees to better assess the health risks.

Without any safety measures, 50% of the employees face the risk of being severely impacted by the virus, with 16 possible deaths.
Race-related risks project that a total number of 49 or more people could be infected, having 10 hospitalizations and 2 deaths as a result.
Gender-related risks may cause 53 or more people to get infected, 10 of these people to be hospitalized, and 4 people to die.

After the risk assessment, we determined the groups who are most at risk at Bellevue College are:

- People older than 60 (300-500 people total),
- Minorities (specifically, Native Hawaiian, Pacific Islanders, Asians and Hispanics(about 4000 people); possibly about 37 people infected, 7 people hospitalized, 1 person dead).

It is important to mention that all people are to some extent at risk from COVID-19.

Task 3: Identify how healthy the students and campus employees are, compared to the general population.

After conducting extensive research, comparing statistics, and analyzing data, we discovered that the general population of King County is healthy compared to the rest of Seattle and even Washington. Nearly 80% of the population is healthy and doesn't have any underlying conditions that would make COVID-19 especially deadly.

Bellevue College has a higher percentage of a healthy population, at 92.31%. This means around 8% of Bellevue College's community is at an increased risk of contracting a severe or life-threatening case of COVID-19. In conclusion, we have determined that Bellevue College has a better state of health than the general public.

Task 4: Find out whether schools have a higher transmission rate than general workplaces and community transmission in general.

Bellevue College's administration should consider whether to reopen its campus in fall because school settings could have a higher transmission rate than other general workplaces. Although some schools opened with relatively little impact on the transmission rate, we don't know whether the same would be true for higher education schools, where students are older and more likely to be affected by the disease and transmit the virus.

Task 5: Determine what will happen to classroom instruction if professors get sick and how easily replaceable the professors are.

Bellevue College faces a challenging decision when considering the effects that

could occur to professors and staff falling ill with COVID-19. While many professors within the same department can substitute for their fellow professors, more than a few professors being unable to teach could dramatically affect the education of students and the availability of instruction that many students rely on.

Bellevue College professors are not easily replaceable and cannot replace professors within other departments. Due to budget cuts, the amount of potential professors on staff is limited and there will not be new professors being onboarded within the near future. Ultimately, putting professors at risk can endanger both the students and staff as well as have a significant impact on instruction and education.

Task 6: Determine how comfortable students and faculty are coming back to live classrooms with PPE (personal protective equipment and other infection reductions measures we would define through research) and without PPE.

Based on the data we received from our survey, we have concluded that the majority of the people who responded are comfortable with attending on-campus classes, wearing a mask, and taking precautions to reduce infections. We have also concluded that for the most part, people do not have a medical condition that could higher the risk of them getting COVID-19, nor do they live with or visit somebody who has such a condition.

For the most part, the majority of the respondents are not in any situation where they might be at risk of getting COVID-19. More than half of the respondents did not have or live with/visit someone that has a medical condition that made them have a higher risk of getting COVID-19. This clearly shows that most of the people are not in situations in which they are at risk of getting COVID-19.

Given that social distancing guidelines are followed, 6 people (11.54%) are uncomfortable going to campus and 7 people (13.46%) are not going to attend on-campus classes at all. On the other hand, the remaining 39 people are either somewhat comfortable, comfortable or not worried at all about attending on-campus classes. As seen from this data, the majority of the people are comfortable to attend on-campus classes.

As for how comfortable with how people are wearing masks, only 2 people (3.85%) are unwilling to wear masks, whereas the rest of the people are willing to wear masks, all of which have mixed feelings about doing so. The remaining people either think it will impact their learning, would prefer not to wear them, are willing to wear them, and go to places where people do not wear them, or they

believe that masks must absolutely be worn. This data clearly shows that the majority of the people are willing to wear masks, even if some of them would rather not.

Regarding safety precautions to ensure that the virus does not get spread around, 48 people (92.31%) are willing to help sanitize their workplaces before and after their use and only 4 people (7.69%) are not willing to do so. It can be concluded from this data that most people are willing to make sure that the risk of infection is reduced as much as possible.

Task 7: Determine whether students prefer online or on-campus learning.

A quarter of the students and staff who participated in our survey answered that they would not come to on-campus learning or would be uncomfortable doing so. Another 25% responded that they would be somewhat comfortable, but they were worried about the impact their concern for contracting the virus could have on their education. The rest of the surveyees (50%), stated that they were comfortable, or not worried at all.

Students were also asked if they learn well online, to which 27 surveyees, or 51.93%, responded that they're either good with online classes, or prefer them. However, 14 surveyees(26.92%) stated that though online classes were alright, they would do better at live classes, while the rest of the 21.26% believed that they would either fail or not get proper education from online classes.

Task 8: Determine technology that can help the campus reduce class sizes, so it can maintain social distancing. Brainstorm any other computer-based programs or ideas that could allow the college to reduce the number of students in class and improve online instruction.

Bellevue College is currently using ZOOM, Microsoft Teams, and ConexED for online instructions. After a substantial amount of research was done, we found that there is a host of software that share similar features with the software that BC is using and some of them provide additional features that are beneficial to online learning.

Classes at Bellevue College that are using ZOOM as their primary platform for online instructions could consider switching to other software that we discovered and included in our report due to the ongoing security issue with ZOOM.

We looked at software, such as Google Meet, discord, Webex, etc. and listed features that each provides. We agreed that Google Meet was the ideal option

due to its stability and security. With Google Meet, students can feel safe learning and don't need to constantly worry about their personal information being exposed or obtained by strangers. Additionally, Google Meet provides several means of access. For instance, users can join a call through Gmail, Google calendar, and Google slides. The background noise filtering feature that Google Meet offers is also great because it automatically removes distracting sound accidentally picked up by students' microphones during the class.

Task 9: Determine what equipment and programs we would need to live stream classes to students at home.

Bellevue College currently has computers with 8MB hard drives for each classroom and it plans to provide student computers to students who do not have access to a computer. However, not every professor has a camera or microphone system yet. Additionally, due to the coronavirus outbreak, many internet services such as Comcast are offering their internet services for free, so students without internet access will be provided with free internet.

Using this information, as well as some research, we have concluded that the college not only needs to purchase cameras or microphone systems, but it has to purchase capture cards, video encoders, and possibly new laptops/desktops as well.

There are various reasons as to why we concluded that this equipment is necessary. Camera or microphone systems, of course, are necessary for the students to hear, see and communicate effectively with the teacher during lectures, which is invaluable when teachers are giving a hands-on demonstration of some sort. Some cameras require a capture card or they will not even work, so the purchase of a capture card may be necessary depending on what camera is bought. Video encoders can compress videos while they are being streamed, meaning that lag and low-quality videos will not be a problem for students. Because lag and low-quality videos can negatively affect online learning, video encoders are necessary. As for laptops/desktops, although Bellevue College has Dell computers with 8MB hard drives for each classroom, that may not be enough for some students, as they may require certain specs. Not much information was given about the specs of each computer, so purchasing new laptops/desktops may be necessary.

Task 10: Determine if the campus needs to upgrade and/or increase its servers and supporting software.

Based on the best available information, Bellevue college is currently using 173 virtual servers maintained by a cloud provider and 85 servers on-premise. It is possible that shifting classes remotely increased the traffic to all servers, but because the majority of the servers are in the cloud and most likely have the ability to auto-scale, the increase in traffic does not present any major concerns at this point. However, Bellevue College should consider whether to move all of its servers in the cloud, as this could prevent any server overload and infrastructure failure.

Task 11: Determine when (and if) faster internet technologies such as 5G or 4G may come online in the US, which could help with online instruction possibilities. Determine whether faster internet speeds are available or may become available till fall.

While there is a small percentage of households in the Seattle area who are not connected to the Internet, the vast majority of the area is connected. 4G has been available across Washington State for almost a decade, and while it may not be the fastest available option, it is stable enough for basic online instruction activities. For those who do not have wired or wireless internet options available in their households, there are numerous locations such as schools, libraries, and coffee shops where internet access is available.

Task 12: Determine what is the current method of tracking Coronavirus cases among Bellevue College community and research new ways of facilitating the task if needed.

Currently, Bellevue College is planning to track COVID-19 infections in the Bellevue College community through manual contact tracing, mandatory health questionnaires, mandatory temperature readings, and self-reporting. The methods of Bellevue College are largely dictated by Washington State in the Phase 1 Higher Education & Critical Infrastructure Workforce Training Restart COVID-19 Requirements.

A technology that could better facilitate the efforts of Bellevue College to track COVID-19 infections in its community is digital contact tracing. Digital contact tracing is a technology that uses the electronic data of individuals along with

reports of COVID-19 infections to inform health officials of who may have come into contact with an infected person.

The most notable program that has been developed to facilitate digitally contact tracing is an API developed jointly by Apple and Google, that health officials could build a digital contact tracing from. Apple and Google are also developing opt-in software that would digitally contact traces through gathering data on Bluetooth interaction between individuals.

Though digital contact tracing would likely help Bellevue College's efforts to track COVID-19 infections, the programs are controversial due to privacy and legality concerns. Furthermore, digital contact tracing programs are almost exclusively aimed at adoption by government officials. Thus, despite possibly benefiting from digital contact tracing programs, it appears unlikely that Bellevue College could directly adopt them.

Recommendations

Below we present our recommendations related to each individual task.

Task 1: Acquire an understanding of what the health risks for campus employees and students are, based on the best, currently available information.

As new studies related to the virus emerge often and the situation evolves rapidly, we recommend that Bellevue College continues to stay up to date with the newest information about the pandemic in order to make the best decisions. The campus should follow state directives and health officials' recommendations regarding social distancing measures.

Task 2: Obtain data on the demographics of the students and campus employees to better assess the health risks.

We believe that Bellevue College's administration should focus on the protection of high-risk individuals. High-risk individuals are people older than 60 and members of certain racial minority groups (specifically, Native Hawaiian, Pacific Islanders, Asians, and Hispanics). Other groups should be protected as much as

possible because, without any measures, the campus endangers the health of a number of people.

Task 3: Identify how healthy the students and campus employees are, compared to the general population.

We recommend that Bellevue College starts a program that can successfully determine the general state of the health of the students and staff. There are many ways to achieve this. For example, the administration could do a recurring survey to collect data and input into a format of a website link or email. Another recommendation is to implement something that would inform the administration of someone's health situation, so other services could be provided to ensure the safety of Bellevue College's students and employees.

Task 4: Find out whether schools have a higher transmission rate than general workplaces and community transmission in general.

Bellevue College's administration should consider whether to reopen its campus in fall because school settings could have a higher transmission rate than other general workplaces. Although some schools opened with relatively little impact on the transmission rate, we don't know whether the same would be true for higher education schools, where students are older and more likely to be affected by the disease and transmit the virus. We recommend that Bellevue College keeps online classes or hybrid classes format for as long as possible, instead of going back to the regular classes.

Task 5: Determine what will happen to classroom instruction if professors get sick and how easily replaceable the professors are.

Due to lack of budget and the limitations on available professors, we recommend that Bellevue College's administration should strongly consider continuing remote education until state directives deem that it is safe. If a professor were to become infected it would have a severe effect on the department and can become exponentially worse if other professors were to fall ill as well.

Task 6: Determine how comfortable students and faculty are coming back to live classrooms with PPE (personal protective equipment and other infection reductions measures we would define through research) and without PPE.

Based on the responses of the 52 participants in our survey, there are some things we recommend the college to do. To begin, we recommend that the college does allow on-campus classes to take place. The reason why we believe this is because according to the data received from the surveys, not only are the majority of the people comfortable with taking on-campus classes, most of them are also not in any situation where they are at risk of getting COVID-19. For the people that choose not to take any risks, they can still take all online classes. Additionally, we would like to recommend that the college enforces people to wear masks, as the data suggests that the majority of the people are comfortable with wearing them, with some even believing that they should be necessary. Finally, we would recommend that the college makes sure the necessary steps are being taken in order to reduce the risk of infections. We believe this should happen because the majority of the respondents on the surveys stated that they would be willing to sanitize their desks and workspaces before and after they use them. If the college takes these courses of action, not only will the staff and students be mostly satisfied, they will also be safe and able to learn efficiently as well.

Task 7: Determine whether students prefer online or on-campus learning.

Following the responses of the student surveyees, we recommend that the campus obligates students and employees to wear face coverings since most people are confident in coming to on-campus classes if everyone is required to wear a mask or a face covering. Another recommendation would be to put reminders for social distancing at doors, building entrances, common meeting places, etc., to ensure the health and build the confidence of students and staff. Finally, another recommendation we have is to keep sanitizers and cleaning products to sanitize commonly touched places, such as doors, stair rails, and tables, and to have a sanitizer dispenser in classrooms to stem the spread of germs.

Task 8: Determine technology that can help the campus reduce class sizes, so it can maintain social distancing. Brainstorm any other computer-based programs or ideas that could allow the college to reduce the number of students in class and improve online instruction.

It has been a few months since Bellevue College adopted online instructions. We believe that most faculty and students have grown accustomed to using computer software for learning, but in order to continue improving students' education quality and learning environment, it is recommended to try other computer software that offers different features that could help students learn better and more efficiently.

Task 9: Determine what equipment and programs we would need to live stream classes to students at home.

After doing our research and determining which items were needed specifically, we recommend that Bellevue College purchases the following items for every classroom:

- Cameras:
    - Any of the following:
        - Canon Vixia HF R400 ($200)
        - Canon VIXIA HF R800 (under $300)
- Capture Card:
    - Blackmagic Design Intensity Pro ($190)
- Microphones:
    - Any of the following:
        - Audio Technica ATR2100 ($60)
        - Pyle-Pro PDMIC58 ($13)
        - Shure PG185 Lapel Mic and Wireless System ($299)
- Video Encoders:
    - Any of the following:
        - VidiU ($699)
        - LiveShell X ($699)
- Laptops/Desktops:
    - Any Mac or PC with the following specs (at least $600):
        - High definition display: 1280 x 720
        - i5 Intel Processor
        - 8GB RAM

- 1GB Video RAM

Task 10: Determine if the campus needs to upgrade and/or increase its servers and supporting software.

Although most of the servers are maintained by cloud providers, and most likely can auto-scale easily, we recommend that Bellevue College evaluates its servers and verifies the servers and ensures that their capacity is not exceeded by the increased demand in online services. We also recommend, if the college affords, that BC considers moving all its servers in the cloud to ensure an automatic increase in capacity on a need basis. It is possible that transferring all the servers in the cloud would result in a cost reduction. Maintaining servers on-premise could be more expensive than having them maintained by a cloud provider, as it requires buying additional equipment and hiring employees for system administration. A cloud provider would also be able to provide fail-over support, which could prevent data loss. Moving the servers with a cloud provider would also mean fewer employees on campus, who are likely to increase the chance of spreading the virus among the Bellevue College community.

Task 11: Determine when (and if) faster internet technologies such as 5G or 4G may come online in the US, which could help with online instruction possibilities. Determine whether faster internet speeds are available or may become available till fall.

The data suggests that Seattle is being considered for 5G deployment, but no available timeline exists as to when 5G will be available anywhere in the greater Seattle area. We recommend that Bellevue College continues to use its current internet provider. Bellevue College should instead focus its resources on assisting the students who do not have access to reliable internet.

Task 12: Determine what is the current method of tracking Coronavirus cases among Bellevue College community and research new ways of facilitating the task if needed.

Considering the advantages that digital contact tracing could provide Bellevue College in its effort to track COVID-19 infections, we recommend that Bellevue College officials ask State and County officials to adopt digital contact tracing technology.

References

Task 1 References

Alltucker, K. (2020, May 3). Antibody tests were supposed to help guide reopening plans. They've brought more confusion than clarity. Retrieved from https://www.usatoday.com/story/news/health/2020/04/30/coronavirus-antibody-tests-immunity-mayo-clinic-who/3038740001/

The article contains information about the inaccuracy of the antibody tests developed by some companies. It also addresses the danger of relying on the antibody testing for making policies and decisions regarding who is eligible for staying out of lockdown. Scientists warn that there is no proof that those who recover from COVID-19 will develop antibodies or whether they will be immune from a follow-up infection.

Brosseau, L. (2020, March 16). COMMENTARY: COVID-19 transmission messages should hinge on science. Retrieved from https://www.cidrap.umn.edu/news-perspective/2020/03/commentary-covid-19-transmission-messages-should-hinge-science

The article considers the possibility of Covid-19 being able to transmit through aerosols, tiny particles that hang in the air long after an infected person has coughed or sneezed. These aerosols could subsequently be inhaled and infect people even at a distance greater than the currently recommended 6 feet.

Coronavirus (COVID-19) frequently asked questions. (2020, April 22). *CDC*. Retrieved from https://www.cdc.gov/coronavirus/2019-ncov/faq.html#How-COVID-19-Spreads

This is the CDC Q&A section that answers questions regarding the new COVID-19 disease.

COVID-19: data. (2020). *NYC Health*. Retrieved from https://www1.nyc.gov/site/doh/covid/covid-19-data.page

This is the NY department of health, COVID-19 section.

Drew, J. (n.d.). Elective surgeries, dental services can resume, Inslee announces. Retrieved from https://www.thenewstribune.com/news/coronavirus/article242821251.html

The article informs about Governor Inslee's decision to reopen dental services to Washington State residents. It also emphasizes the guidelines under which dental offices are allowed to operate. The article also elaborates on the criteria to be met before counties are ready to move to Phase 2 of his reopening plan.

Doughton, S. (2020, May 28). Half of the newly diagnosed coronavirus cases in Washington are people under 40. Retrieved from https://www.seattletimes.com/seattle-news/health/half-of-newly-diagnosed-coronavirus-cases-in-washington-are-in-people-under-40/

The article reports recent analysis results about the impact of the new coronavirus on different age groups. It informs that more cases are found in people younger than 40, as opposed to previous reports that stated that more than two-thirds of cases occurred in older people, usually more than 60 years old. The article also ponders over the idea of building herd immunity to overcome the epidemic. Herd immunity is a term used for when enough population has become immune to the virus and as a result, the transmission rate falls naturally. Experts warn that herd immunity is typically built through vaccination and that for the disease to stop spreading naturally, 70% of the population would have to be infected, a percentage that is highly dangerous as it would overwhelm the healthcare system and would increase the chance of death for those left without medical care. The latest results based on a modeling show that there are currently less than 2% of people that have been infected so far in King County.

Klein, S. (2020, April 17). At-home 'swab and scan' kits identify 44 COVID-19 cases in King County. *My Northwest*. Retrieved from https://mynorthwest.com/1822510/home-swab-scan-kits-covid-cases-king-county/

Kottasova, I., & Pettersson, H. (2020, March 20). Data from China shows the majority of people with COVID-19 only suffer mild symptoms, then recover. *CNN Health*. Retrieved from https://www.cnn.com/2020/03/20/health/covid-19-recovery-rates-intl/index.html

Lutton, L. (2020, April 24). Coronavirus case numbers in the United States: APRIL 24 update. *Medical Economics'*. Retrieved from https://www.medicaleconomics.com/news/coronavirus-case-numbers-united-states-april-24-update

McCammon, S. (2020, May 16). 13 USS Roosevelt Sailors Test Positive For COVID-19, Again. Retrieved from https://www.npr.org/sections/coronavirus-live-updates/2020/05/16/857379338/5-uss-roosevelt-sailors-test-positive-for-covid-19-again

The article describes the situation created by COVID-19 on the Theodore Roosevelt ship. It warns about 12 cases of reinfection with COVID-19 after the sailors have supposedly completely recovered. The article elaborates on the possible cause for retesting positive in the sailors' case, concluding that the most likely scenario is that the virus remains dormant in some patients.

Public Health Insider. (2020, May 18). Media Release: Face covering directive goes into effect today: Prevent spreading COVID-19 when unable to maintain 6 feet of distance. Retrieved from https://publichealthinsider.com/2020/05/18/media-release-face-covering-directive-goes-into-effect-today-prevent-spreading-covid-19-when-unable-to-maintain-6-feet-of-distance/

The article informs about governor Inslee's new directive regarding face coverings. The directive requires most of the general population to wear a face-covering when indoor in public places and outdoors when it is not possible to keep a distance bigger than 6 feet. This will prevent infectious respiratory droplets to travel and transmit to other people.

Public Health Insider. (2020, May 14). Media release: Testing recommended for anyone with mild COVID-19 symptoms; decreased cases in long-term care facilities shows importance of aggressive response. Retrieved from https://publichealthinsider.com/2020/05/13/media-release-testing-recommended-for-anyone-with-mild-covid-19-symptoms-decreased-cases-in-long-term-care-facilities-shows-importance-of-aggressive-response/

The article brings people's attention to the new broadened criteria under which patients qualify to test for COVID-19. It informs that anyone with mild symptoms or who has been in contact with an infected person should test for coronavirus, as opposed to previous guidelines, which, due to a shortage of supplies, advised that these people were not tested.

Public Health Insider. (2020, May 8). Media Release: Latest report shows COVID-19 transmission rate could be rising in King County; social distancing remains key. Retrieved from https://publichealthinsider.com/2020/05/08/media-release-latest-report-shows-covid-19-transmission-rate-could-be-rising-in-king-county-social-distancing-remains

The article reports the latest data regarding transmission rates in King County and concludes that, as of April 6, the cases could be rising again after a previous indication that the infection rate was falling.

Q&A: Influenza and COVID-19 - similarities and differences. (n.d.). Retrieved from https://www.who.int/emergencies/diseases/novel-coronavirus-2019/question-and-answers-hub/q-a-detail/q-a-similarities-and-differences-covid-19-and-influenza?gclid=CjwKCAjwtqj2BRBYEiwAqfzur_tziUQloJZR7Unbpzwr453HdEUf0sn-f5ivjMiAfabGETpzmXap-RoCUMUQAvD_BwE

The article answers questions regarding the differences between coronavirus and seasonal influenza. It addresses aspects such as the speed of transmission, incubation period, the severity caused by each virus in different people, mortality rate, vaccines, and available therapeutics.

Quinn, P., & KOMO News. (2020, May 5). New UW model shows COVID-19 nationwide death toll to reach 134,475 by early August. Retrieved from https://komonews.com/news/coronavirus/new-uw-model-shows-covid-19-nationwide-death-toll-to-reach-134475-by-early-august

This article describes experts' new projections of the COVID-19 impact for the summer. The data extracted from the models made by experts from the Institute of Health Metrics and Evaluation at the University of Washington shows a sobering new estimate for the summer and concludes that the impact will be greater than previously thought. This is due to the increased mobility because of the relaxed lockdown measures.

Quinn, M. (2020, May 21). U.S. commits $1.2 billion to production of Oxford coronavirus vaccine. Retrieved from https://www.cbsnews.com/news/oxford-coronavirus-vaccine-oxford-astrazenca-partnering-united-states/

This article informs that the Trump administration has launched Operation Warp Speed and has partnered with AstraZeneca to expedite COVID-19

vaccine development. It also committed 1.2 billion US dollars in order to obtain about 300 million vaccines by the end of the year.

Radcliffe, S. (2020, May 18). Where We're at with Vaccines and Treatments for COVID-19. Retrieved from https://www.healthline.com/health-news/heres-exactly-where-were-at-with-vaccines-and-treatments-for-covid-19#Clinical-trial-stages

This article is summarizing information regarding treatment aspects of COVID-19, such as vaccine development, and available drugs. It enumerates the companies that are currently in the first phase of the vaccine development and states their approach to making the vaccine. It also names the drugs that are currently being screened for possible effects against COVID-19, as well as categorizes them by the way they act on the virus and how it helps the immune system.

Neergaard, L. (2020, May 15). US begins 'warp speed' vaccine push as studies ramp up. Retrieved from https://apnews.com/756e5b743058701c4a2ebefd0af1ade4

This article provides information related to Operation Warp Speed. It also touches on the timeline for vaccine development. The article summarizes Trump's plans, as well as NIH's plan for how they will maximize their efforts to speed up vaccine development.

Sarah Kaplan, W. W. (2020, March 23). The coronavirus isn't alive. That's why it's so hard to kill. Retrieved from https://www.washingtonpost.com/health/2020/03/23/coronavirus-isnt-alive-thats-why-its-so-hard-kill/

This article outlines the main containment challenges that coronavirus poses, and describes the reasons. Being able to transmit undetected from people who are unsuspicious of their disease is a significant reason. Replicating in the upper respiratory tract in its early stages is the reason why it is so contagious and explains why people remain mobile and are able to infect more places.

Staff, A. K. I. N. G. 5. (2020, May 18). 'A lot of hope': Experimental Seattle coronavirus vaccine study shows promise. Retrieved from https://www.king5.com/article/news/health/coronavirus/seattle-moderna-covid-19-vaccine-results/281-9b01a0ed-2c1e-42d6-a023-c43ae749c6e8

The article provides updates regarding the progress of Moderna's developed vaccine. It states that the vaccine brings hope and that volunteers who participated in the first testing phase developed antibodies against COVID-19, an immune response that scientists have been hoping for.

Taylor, C. (2020, May 20). Scientists propose a 50 days on, 30 days off coronavirus lockdown strategy. Retrieved from https://www.cnbc.com/2020/05/19/study-proposes-50-days-on-30-days-off-coronavirus-lockdown-strategy.html

The article presents a group of experts' study and the data from models running different scenarios. The study concluded that the best approach to COVID-19 related lockdown measures is a plan in which lockdown measures are strictly imposed for 50 days, followed by 30 days of more relaxed measures.

Test for Past Infection (Antibody Test). (2020, May 23). Retrieved from https://www.cdc.gov/coronavirus/2019-ncov/testing/serology-overview.html

This article provides CDC's information regarding the antibody tests, what the general public should know when taking the test, and how to interpret the results. It also explains the current situation created around tests developed by various companies and their inaccuracies.

Woodward, A. (2020, May 14). Anthony Fauci says there's 'virtually no chance' the coronavirus will disappear - 3 charts predict what future waves of cases will look like. Retrieved from https://www.businessinsider.com/second-wave-of-coronavirus-infections-may-peak-in-fall-2020-5

This article presents projections of the future coronavirus waves. It describes 3 scenarios that could happen in the next 18-24 months. The first scenario involves many medium waves happening at regular intervals during the next 18 months. The second and the worst scenario, which Dr. Faucci believes is inevitable, would be the result of prematurely relaxed social distancing measures and involves a much greater wave than the one the country has seen already, happening in fall and winter. The second scenario would require lockdowns measures imposed again. The third scenario is the best scenario and involves a "slow burn" of new infections for the next 18-24 months. Although experts believe this

scenario is unlikely, it remains a possibility. No lockdown measures would be necessary for this scenario.

Woodward, A., & Gal, S. (2020, April 7). One chart shows how long the coronavirus lives on surfaces like cardboard, plastic, wood, and steel. *BUSINESS INSIDER*. Retrieved from https://www.businessinsider.com/coronavirus-lifespan-on-surfaces-graphic-2020-3

The article discusses the viability of the coronavirus on different surfaces. It's viability depends on the surface and ranges between three hours to 7 days.

Task 2 References

BC Facts at a Glance (n.d.). Retrieved May 30, 2020, from https://36d5l8225ig13rrnnc3w4af9-wpengine.netdna-ssl.com/wp-content/uploads/sites/108/2017/11/Facts-At-A-Glance-Booklet-8-17.pdf
This booklet provides partial demographics data for the population of BC (2017)
(Does not have age for faculty members nor does it have the gender distribution for staff)

Bellevue College Student Population and Demographics. (n.d.). Retrieved May 30, 2020, from https://www.univstats.com/colleges/bellevue-college/student-population
This article provides full demographics data for students of BC (2017-2018)

Task 3 References

Gov., K.C. (2020, April 16). Community health indicators. Retrieved May 27, 2020, from https://www.kingcounty.gov/depts/health/data/community-health-indicators.aspx

This article provides data and numerical statistics pertaining to the general health and state of King County. It provides a full overview of the health and demographics of King County residents.

Gov., WDOH. (n.d.). Community and Environment. Retrieved May 27, 2020, from https://www.doh.wa.gov/CommunityandEnvironment/ClimateandHealth

This article elaborates on general climate information and climate change in Washington. It gives information and data on aspects such as air quality, water cleanliness, agriculture, etc., and how these affect our health as a community.

U.S. Cancer Statistics Working Group. U.S. Cancer Statistics Data Visualizations Tool, based on November 2018 submission data (1999-2016): U.S. Department of Health and Human Services, Centers for Disease Control and Prevention and National Cancer Institute; www.cdc.gov/cancer/dataviz, June 2019. Retrieved May 27, 2020, from https://gis.cdc.gov/Cancer/USCS/DataViz.html

This article shows the data on the cases, deaths, and types of cancer in the entire United States. It gives information on the cancer status of states and cities. The article shows the prevalence of cancer, which types, and the death rate according to the type and area.

Gov., King County. (2019, October 31). King County City Health Profiles. Retrieved May 27, 2020, from https://www.kingcounty.gov/depts/health/data/city-health-profiles.aspx

This article shows an overview of the demographics of King County. It has intro, cities, reports, and ranking tabs to explore, learn, and compare different areas and download a full report with demographics for an area. It also compares neighborhood ranking, region, and topic through the ranking tab.

Gov., CDC. (2020, March 24). Most Recent Asthma State Data. Retrieved May 27, 2020, from https://www.cdc.gov/asthma/most_recent_data_states.htm

This article presents data on asthma in the United States of America. It shows the adult prevalence of asthma by state, along with the exact number and percentages. It also has a tab for the number of deaths and the mortality rate.

Task 4 References

Deliso, M. (2020, May 27). How schools around the world are reopening during the coronavirus pandemic. *ABC News*. Retrieved from https://abcnews.go.com/International/schools-world-reopening-coronavirus-pandemic/story?id=70641371

The article depicts other countries, such as China and Denmark, on the methods they are using to safely reopen their schools. They also provide the different strategies of teaching their faculty is implementing during this time.

Boffey, D., & Willsher, K. (2020, May 18). Schools reopening has not triggered a rise in COVID-19 cases, EU ministers told. *The Guardian*. Retrieved from https://www.theguardian.com/world/2020/may/18/french-minister-tells-of-risks-of-missing-school-as-more-pupils-return-covid-19

The article considers the positive opinion of reopening schools and shows examples of reopening schools in the EU. However, they also consider the possibility of this increasing the spreading of coronavirus through schools.

Brosseau, L. (2020, March 16). Commentary: COVID-19 transmission message should hinge on science. *CIDRAP*. Retrieved from https://www.cidrap.umn.edu/news-perspective/2020/03/commentary-covid-19-transmission-messages-should-hinge-science

The article shows how dangerous for close-range aerosol transmission in small spaces such as classrooms in the school, and intuitive illustrations used helps us to be understandable. It is recommended that we protect ourselves by wearing face coverings.

Buchholz, K. (2020, May 13). One infected clubgoer triggers massive response. Statista. Retrieved from https://www.statista.com/chart/21697/contact-tracing-response-seoul-south-korea-covid-19/?utm_source=Statista+Global&utm_campaign=c160f956fc-All_InfographTicker_daily_COM_AM_KW20_2020_Fr&utm_medium=email&utm_term=0_afecd219f5-c160f956fc-304569477

The article shows how to affect a person who has a positive result on the COVID-19 can spread rapidly and increase the number of positive people in the various areas in many routes using the example in Korea. It is also up-to-date, and it indicates which points we should consider.

Christakis, D. (2020, May 13). School reopening-the pandemic issue that is not getting its due. *JAMA Pediatrics*. Retrieved from https://jamanetwork.com/journals/jamapediatrics/fullarticle/2766113

The article explains schools have been closed down because any virus spreads at a faster speed through a child rather than an adult. It also provides detail on how people are worried about the education their children are receiving in this period of online meetings and lectures.

EClinical Medicine. (2020, April 28). What are the underlying transmission patterns of COVID-19 outbreak? An age-specific social contact characterization. *THE LANCET*. Retrieved from https://www.thelancet.com/journals/eclinm/article/PIIS2589-5370(20)30098-5/fulltext

The article explains the geographical locations and the disease situations, as shown in 6 major cities in China where are known as severe results related to COVID-19. From the data, we can refer to how to cause the special where it is close geographically because the circumstance of classrooms of school is similar.

Henriques-Gomes, L. (2020, April 26). Children unlikely to transmit coronavirus, says study cited in PM's push to reopen schools. *The Guardian*. Retrieved from https://www.theguardian.com/world/2020/apr/26/children-unlikely-to-transmit-coronavirus-says-study-cited-in-pms-push-to-reopen-schools

This article summarizes how schools are being pushed to reopen since children are less likely to transmit the virus. Many studies have been done by the NSW Health's Centre for Immunization Research on kids examining the spread of the Coronavirus from 15 different schools.

Gong, S. (2020, May 12). South Korea delays reopening schools as Coronavirus cases resurge. *NPR*. Retrieved from https://www.npr.org/sections/coronavirus-live-updates/2020/05/12/854407305/south-korea-delays-reopening-schools-as-coronavirus-cases-resurge

The article offers a warning that reopening schools may increase the number of people who are transmitting COVID-19, with high speed. This is

just the example that happened last week in Korea, and it is still going on a serious issue whether they would keep reopening schools or not.

LaFaro, A. (2020, April 24). Household deemed most common place for COVID-19 transmission. *UNC*. Retrieved from https://research.unc.edu/2020/04/24/household-deemed-most-common-place-for-covid-19-transmission/

The article considers that the same household is more dangerous than outside places if there is one person who is infected with COVID-19, and it can be the place to start.

Maresca, T. (2020, May 8). South Korea sees a new cluster of COVID-19 cases tied to nightclubs. *UPI*. Retrieved from https://www.upi.com/Top_News/World-News/2020/05/08/South-Korea-sees-new-cluster-of-COVID-19-cases-tied-to-nightclubs/5491588933169/

The article considers how a new cluster of COVID-19 cases is tied to one infected person who visited various nightclubs. The incident also caused schools to postpone reopening. They believe they will have another community infection due to this issue. Scientists think transmission due to unknown sources of infection can lead to widespread clusters of infection.

Moreno, Y. (2011, December 29). Controlling the spread of disease in schools. PMC. Retrieved from https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3248433/

This article focuses on controlling the spread of coronavirus in schools. The article also states that public health campaigns educated people about protective practices, such as wearing masks and social distancing. The article also talks about some graphs depicting how much the virus can spread when one wears protective equipment and doesn't.

Task 5 References

Professor Substitution [Email to K. Peterson]. (2020, May 16).

The email from Professor Peterson outlines Bellevue College's policies on time off and sick leave. In the email, Professor Peterson also discusses the potential effects that losing multiple professors can have on a department as well as the difficulty in finding substitute professors.

Task 8 References

ConexED Reviews and Pricing - 2020. (n.d.). Retrieved from https://www.capterra.com/p/185157/ConexEd/

This website outlines the features of ConexED and contains reviews from its users, which are helpful in determining whether Bellevue College should continue using this program or switch to other alternative programs.

John, S. (2020, May 18). 'What is Google Meet?': A guide to Google's video-chatting service, including pricing options and how to join a meeting. Retrieved from https://www.businessinsider.com/what-is-google-meet

The article gives a comprehensive guide to Google Meet. It includes information, such as what it is for, the cost of using it, and how to use it to join a meeting call.

Hornshaw, P. (2020, May 8). What Is Discord? Retrieved from https://www.digitaltrends.com/gaming/what-is-discord/

The article contains detailed information regarding Discord, giving step by step instructions in terms of how to use it. It also highlights the features that discord offers.

What is Skype?: Skype Support. (n.d.). Retrieved from https://support.skype.com/en/faq/FA6/what-is-skype

The article provides basic information regarding Skype for first-time users. It also briefly states the additional features that it offers If its users are willing to spend some money.

Adam, B. (2020, May 21). Facebook Rooms, Zoom and Google Meet, which is the best for making video calls? Retrieved from https://intallaght.ie/facebook-rooms-zoom-and-google-meet-which-is-the-best-for-making-video-calls/

The article gathers information about Facebook Messenger, Zoom, and Google Meet and uses that information to have a comparison to figure out which software is most suitable for video chatting.

The Leading Video Platform for Education - Panopto. (n.d.). Retrieved from https://www.panopto.com/panopto-for-education/

This website is panopto's official website. It contains all the information that one needs to know before getting started with the software.

Drake, N., & Turner, B. (2020, May 21). Best video conferencing software in 2020. Retrieved from https://www.techradar.com/best/best-video-conferencing-software

The article looks at 9 different software and gives a brief introduction to each of them. It also includes a link to their respective website for users to gain a better understanding of how each program works.

Brown, L. (2020, April 22). Top 12 Best Webcam Recorder for all Platforms. Retrieved from https://filmora.wondershare.com/screen-recorder/best-webcam-recorder-for-all-platform.html

The article compares 12 most popular webcam recorders and talks about their features and where users can get them. One can click on the name of the webcam recorder that they find interesting and it will direct them to a separate page for more detailed information.

Vigliarolo, B. (2020, April 9). Who has banned Zoom? Google, NASA, and more. Retrieved from https://www.techrepublic.com/article/who-has-banned-zoom-google-nasa-and-more/

The article shows which companies and government agencies are impacted by Zoom's security issues and decided to ban the use of it.

Task 9 References

Barcelos, J. (2018, May 30). Your Essential Equipment Guide for Teaching Online. Retrieved May 31, 2020, from https://www.soulful.mba/blog/essential-equipment-guide

This blog talks about what equipment is needed for online learning, as well

as why and how that equipment is used.

Brown, L. (2020, April 22). Top Live Streaming Hardware Encoder and Equipment you should know. Retrieved May 31, 2020, from https://filmora.wondershare.com/live-streaming/top-live-streaming-hardware-encoder-and-equipment-you-should-know.html

This article lists some options for hardware video encoders, as well as their features and prices.

Cooper, J. (n.d.). How to Go Live with Live Streaming School Video. Retrieved May 26, 2020, from https://www.campussuite.com/blog/how-to-go-live-with-live-streaming-school-video

This blog discusses the essentials of live streaming in a classroom. It also goes into detail about how to produce live streams, live streaming platforms, and sharing live streams.

Managuelod, P. (2020, April 29). 5 Tools For Live Education Broadcasting. Retrieved May 26, 2020, from https://www.dacast.com/blog/live-streaming-classroom-online/

This article talks about the things you need, why you need them, and how to use them when it comes to live streaming for education.

Technology Resources Page :: Public Safety @ Bellevue College. (2020, April 20). Retrieved May 26, 2020, from https://www.bellevuecollege.edu/publicsafety/2020/03/21/technology-resources-page/

This page lists information about software, how to use that software, and free internet services for students.

PDF[https://thejournal.com/~/media/771257FAD08B40E2BB70F3094958B57E.pdf]. (2015, June). Nevada City: Telestream.

This document lists what equipment is necessary for live streaming in a classroom, as well as some options regarding each type of equipment along with their prices.

Task 11 References

Nickelsburg, M. (2019, November 22). As cities battle feds over 5G rollout, tech leaders worry Seattle will lag behind in new wireless race. Retrieved May 28, 2020, from https://www.geekwire.com/2019/cities-battle-feds-5g-rollout-tech-leaders-worry-seattle-will-fall-short-new-digital-divide/

This article provides an overview of where Seattle and Bellevue are in regards to getting 5G coverage. Bellevue and Seattle are both involved in a legal battle with the FCC as to how much cities are allowed to charge the FCC to attach equipment to utility poles.

Looper, C. de. (2020, May 22). Where Is 5G Available?: 5G Network Map. Retrieved May 28, 2020, from https://www.digitaltrends.com/mobile/5g-availability-map/

This article is a compiled list of deployments of 5G by city and state for the major 5G providers in the United States. The article is recent and shows that Seattle and Bellevue do not currently have 5G available.

Broadband in Washington. (2020). Retrieved May 28, 2020, from https://data.wa.gov/stories/s/Broadband-in-Washington/irv9-b275/

This webpage is a compilation of studies regarding broadband Internet in Washington state. It shares the results of the City of Seattle's Technology Access and Adoption Study. The webpage states that ninety-five percent of households in Seattle have wired or wireless access to the Internet.

Frequently Asked Questions: 5G in Seattle. (2019). [PDF file]. Retrieved from http://www.seattle.gov/documents/Departments/Broadband/FAQs_5G_in_Seattle_12.05.19.pdf

This file details the City of Seattle's status in the deployment of 5G. It also gives an overview of the lawsuit against the FCC regarding the prices for placing equipment on utility poles. The file gives a statement that Seattle does not have a timeline for the deployment of 5G but is working towards that goal.

Everything You Need to Know About 5G (2020). Retrieved May 30, 2020, from https://www.qualcomm.com/invention/5g/what-is-5g

This article details the major differences between 4G and 5G. It states 5G will allow for increased speeds and more network traffic, as well as decreased latency between end-users and servers. It also describes the difference between the 4G and 5G spectrums.

Task 12 References

Apple and Google partner on COVID-19 contact tracing technology. (2020, May 20). Retrieved from https://www.apple.com/newsroom/2020/04/apple-and-google-partner-on-covid-19-contact-tracing-technology/

This article is an announcement from Apple and Google that the two companies have partnered to create technology that will help to track COVID-19 infections. The announcement details that two major technologies will be released from the partnership. The first technology is ready for use and is an api that allows governments to create apps and programs that can operate between Apple and Android devices. The second technology is still in development but will be optional software that uses Bluetooth data to track interaction between people.

Center for Disease Control. (2020, March 30). Public health recommendations for community-related exposure. Retrieved from https://www.cdc.gov/coronavirus/2019-ncov/php/public-health-recommendations.html

This page covers the Center for Disease Control's recommendations for mitigating community-related exposure from COVID-19. The page also addresses the issues and limitations of preventing community-related exposure to COVID-19.

Phase 1 Higher Education & Critical Infrastructure Workforce Training Restart COVID-19 Requirements, Phase 1 Higher Education & Critical Infrastructure Workforce Training Restart COVID-19 Requirements (n.d.). Retrieved from https://www.governor.wa.gov/sites/default/files/Phase%201%20Higher%20Education%20COVID-19%20Safety%20Requirements%20Final%201.pdf

This is the Phase 1 Higher Education & Critical Infrastructure Workforce Training Restart COVID-19 Requirements, which details the guidelines that higher education and workforce training organizations must abide by in order to reopen. These guidelines pertain to the first of three planned phases for reopening Washington State.

Phase 1 Higher Education & Critical Infrastructure Workforce Training Restart COVID-19 Requirements :: Public Safety @ Bellevue College. (2020, May 19). Retrieved from https://www.bellevuecollege.edu/publicsafety/phase

-1-higher-education-critical-infrastructure-workforce-training-restart-covid-19-requirements/

This source is from Bellevue College's website and details the Phase 1 Higher Education & Critical Infrastructure Workforce Training Restart COVID-19 Requirements. The page also outlines Bellevue College's plan to address the Phase 1 requirements.

Servick, K. (2020, March 23). Cellphone tracking could help stem the spread of coronavirus. Is privacy the price? Retrieved from https://www.sciencemag.org/news/2020/03/cellphone-tracking-could-help-stem-spread-coronavirus-privacy-price

This source is an article that focuses on the legality, controversy, and implication of using digital contact tracing.

Appendix: Health and learning preferences survey

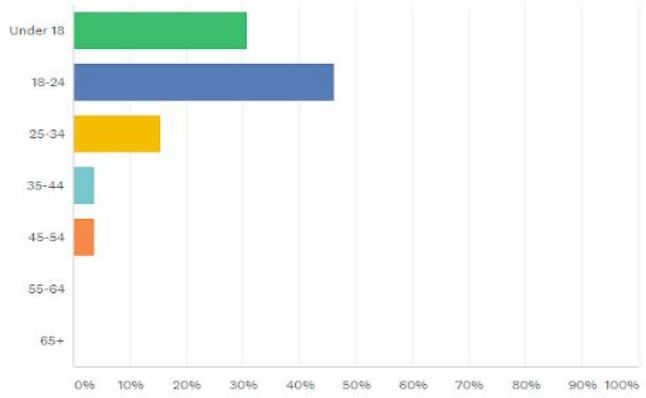
This is the survey we distributed to the 52 students and faculty members.

Q1

Customize Save as

What is your age?

Answered: 52 Skipped: 0



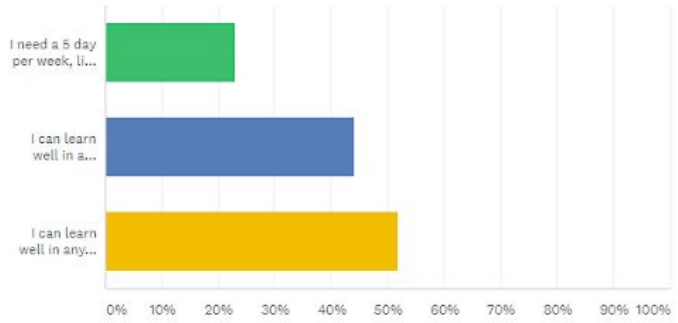
ANSWER CHOICES	RESPONSES	
Under 18	30.77%	16
18-24	46.15%	24
25-34	15.38%	8
35-44	3.85%	2
45-54	3.85%	2
55-64	0.00%	0
65+	0.00%	0
TOTAL		52

Q2

Customize Save as

How do you learn best?

Answered: 52 Skipped: 0



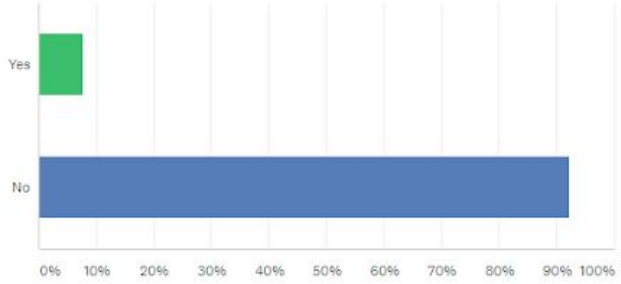
ANSWER CHOICES	RESPONSES	
I need a 5 day per week, live class.	23.08%	12
I can learn well in a hybrid class that meets twice a week and the rest is online.	44.23%	23
I can learn well in any type of class: live, hybrid, or online.	51.92%	27
Total Respondents: 52		

Q3

Customize Save as

Do you have a medical condition that puts you at a higher risk of getting a severe and/or life-threatening case of COVID 19?

Answered: 52 Skipped: 0



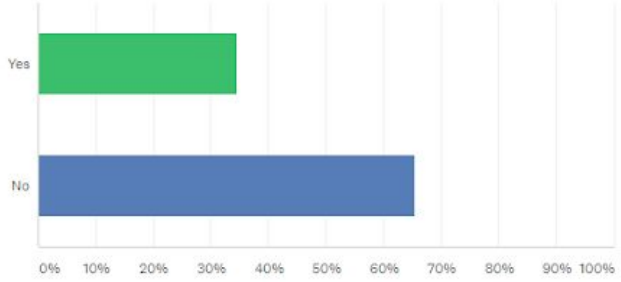
ANSWER CHOICES	RESPONSES	
Yes	7.69%	4
No	92.31%	48
TOTAL		52

Q4

Customize Save as

Do you live with someone who has a medical condition that puts them at a higher risk of getting a severe and/or life-threatening case of COVID 19?

Answered: 52 Skipped: 0



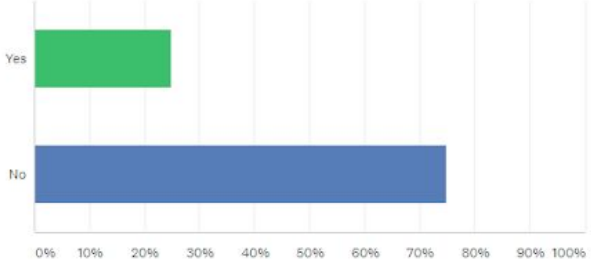
ANSWER CHOICES	RESPONSES	
Yes	34.62%	18
No	65.38%	34
TOTAL		52

Q5

Customize Save as

Do you regularly (at least once a week) visit someone who has a medical condition that puts them at a higher risk of getting a severe and/or life-threatening case of COVID 19?

Answered: 52 Skipped: 0



ANSWER CHOICES	RESPONSES	
Yes	25.00%	13
No	75.00%	39
TOTAL		52

Q6

Customize Save as

Assuming there is no vaccine AND the campus required students and teachers to wear masks, maintain 6 feet of distancing in classrooms, and surveyed/tested for everyone for temperature -- how comfortable would you feel coming back to a live class?

Answered: 52 Skipped: 0

3.3★
average rating

	I WON'T TAKE THE RISK; I'LL TAKE ONLINE CLASSES INSTEAD	UNCOMFORTABLE ENOUGH I WOULD HAVE A HARD TIME LEARNING	SORT OF COMFORTABLE, BUT I MAY FEEL WORRIED DURING CLASS ENOUGH IT COULD IMPACT MY LEARNING	COMFORTABLE ENOUGH I WON'T WORRY MUCH DURING CLASS AND MY LEARNING IS UNLIKELY TO BE IMPACTED	NOT WORRIED AT ALL	TOTAL	WEIGHTED AVERAGE
	13.46% 7	11.54% 6	25.00% 13	28.85% 15	21.15% 11	52	3.33

Q7

Customize Save as

How well do you think you learn online?

Answered: 52 Skipped: 0

3.4★
average rating

	I CAN'T MOTIVATE MYSELF TO WORK IN AN ONLINE CLASS. I WILL FAIL MY CLASSES.	I CAN SORT OF USE ONLINE CLASSES, BUT I DO NOT BELIEVE I AM RECEIVING THE SAME QUALITY OF EDUCATION.	ONLINE CLASSES ARE OK, BUT I DO A BIT BETTER IN LIVE CLASSES.	I AM OK WITH ONLINE CLASSES, AND I DO PRETTY WELL LEARNING IN THEM.	I PREFER ONLINE CLASSES, AND I DO WELL ACADEMICALLY IN THEM.	TOTAL	WEIGHTED AVERAGE
	3.85% 2	17.31% 9	26.92% 14	34.62% 18	17.31% 9	52	3.44

Q8

Customize Save as

Are you willing to wear a mask while you are on campus?

Answered: 52 Skipped: 0

3.9★

average rating

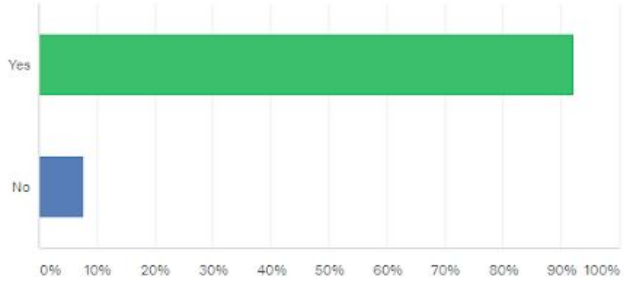
	I WON'T WEAR A MASK. I'LL FIND ANOTHER CAMPUS TO GO TO SCHOOL AT, OR TAKE ONLINE CLASSES.	I'LL WEAR ONE IF I AM FORCED TO, BUT I'LL DISLIKE IT ENOUGH IT COULD IMPACT MY CONCENTRATION.	I AM OK WEARING A MASK, BUT I'D RATHER NOT.	I AM FINE WEARING A MASK, BUT I'M ALSO OK GOING TO PLACES WHERE PEOPLE DON'T WEAR MASKS.	I BELIEVE IT IS CRITICAL FOR ALL OF US TO WEAR MASKS. I FEEL UNCOMFORTABLE AROUND MASK-LESS PEOPLE.	TOTAL	WEIGHTED AVERAGE
☆	3.85% 2	13.46% 7	17.31% 9	21.15% 11	44.23% 23	52	3.88

Q9

Customize Save as

Would you be willing to help sanitize your desks and workstations before and after you use them? (The campus will provide easy to access cleaning materials.)

Answered: 52 Skipped: 0



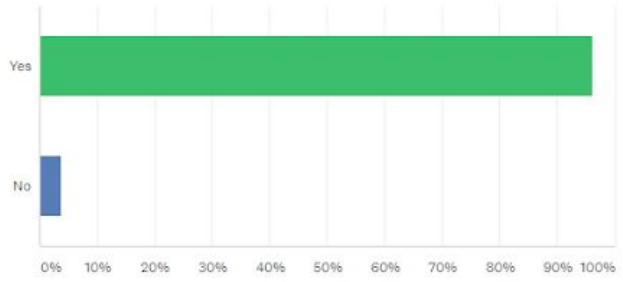
ANSWER CHOICES	RESPONSES	
Yes	92.31%	48
No	7.69%	4
TOTAL		52

Q10

Customize Save as

Do you have access to a computer that works well with online video conferencing at home?

Answered: 52 Skipped: 0



ANSWER CHOICES	RESPONSES	
Yes	96.15%	50
No	3.85%	2
TOTAL		52